US012407304B2

(12) **United States Patent**
Hase et al.

(10) **Patent No.: US 12,407,304 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **POWER AMPLIFIER MODULE**

(71) Applicant: **Murata Manufacturing Co., Ltd.**, Kyoto (JP)

(72) Inventors: **Masatoshi Hase**, Kyoto (JP); **Hitoshi Akamine**, Kyoto (JP)

(73) Assignee: **MURATA MANUFACTURING CO., LTD.**, Kyoto (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 857 days.

(21) Appl. No.: **17/376,780**

(22) Filed: **Jul. 15, 2021**

(65) **Prior Publication Data**

US 2022/0021350 A1 Jan. 20, 2022

(30) **Foreign Application Priority Data**

Jul. 17, 2020 (JP) 2020-122668

(51) **Int. Cl.**
H03F 1/42 (2006.01)
H03F 3/24 (2006.01)

(52) **U.S. Cl.**
CPC ***H03F 1/42*** (2013.01); ***H03F 3/245*** (2013.01); *H03F 2200/171* (2013.01); *H03F 2200/451* (2013.01)

(58) **Field of Classification Search**
CPC H03F 1/42; H03F 3/245; H03F 2200/171; H03F 2200/451; H03F 3/195; H03F 3/68; H03F 3/72; H03F 2203/7209; H03F 3/211; H04B 1/0458; H04B 2001/0408; H04B 1/04
USPC .... 330/295, 124 R
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,130,039 B2 *	3/2012	Dishop	....	H03F 3/211 330/276
8,593,219 B1 *	11/2013	Root	....	H03F 1/0288 330/124 R
2011/0309884 A1	12/2011	Dishop		
2013/0099866 A1 *	4/2013	Conradi	....	H03F 1/0288 330/126
2016/0191105 A1	6/2016	King et al.		
2017/0104456 A1	4/2017	Motoi et al.		

FOREIGN PATENT DOCUMENTS

JP	2016042699 A	3/2016
WO	2015133003 A1	9/2015

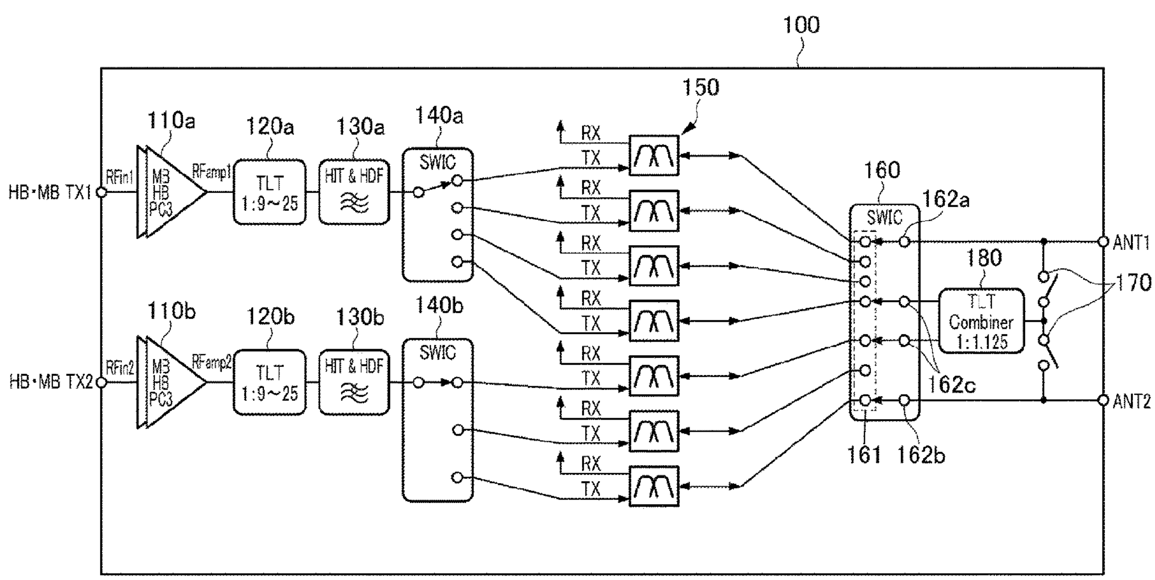
* cited by examiner

Primary Examiner — Andrea Lindgren Baltzell
Assistant Examiner — Khiem D Nguyen
(74) *Attorney, Agent, or Firm* — McDonald Hopkins LLC

(57) **ABSTRACT**

A power amplifier module includes a first amplifier that amplifies a power level of a first input signal in a predetermined frequency band and outputs a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier that amplifies a power level of a second input signal in the predetermined frequency band and outputs a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner that combines the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level and includes a transmission line transformer.

19 Claims, 13 Drawing Sheets



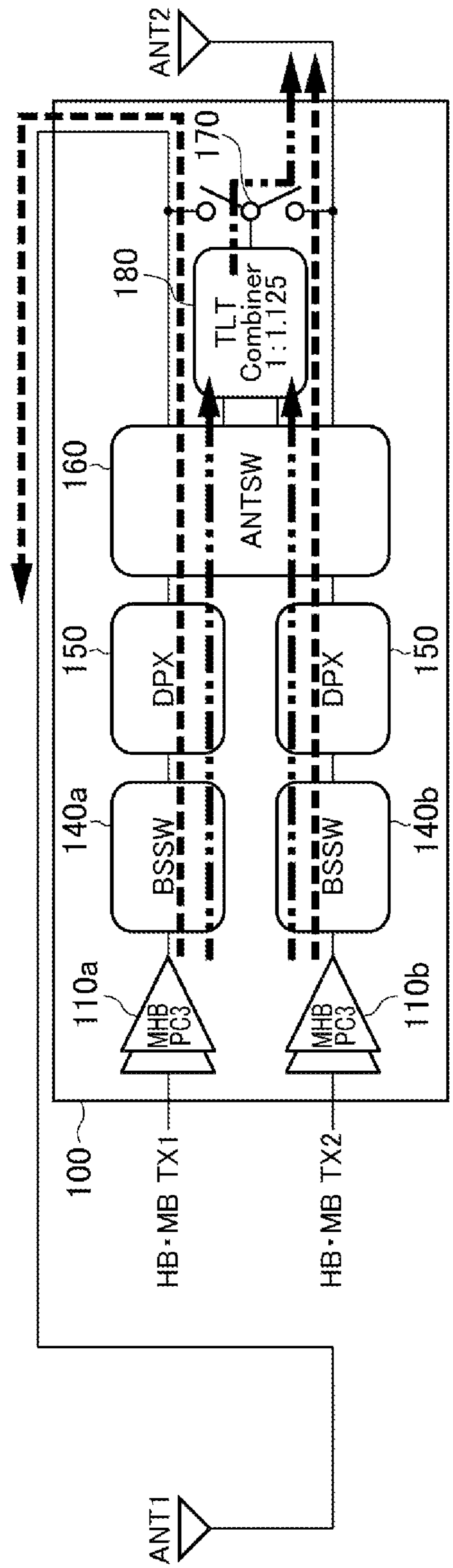
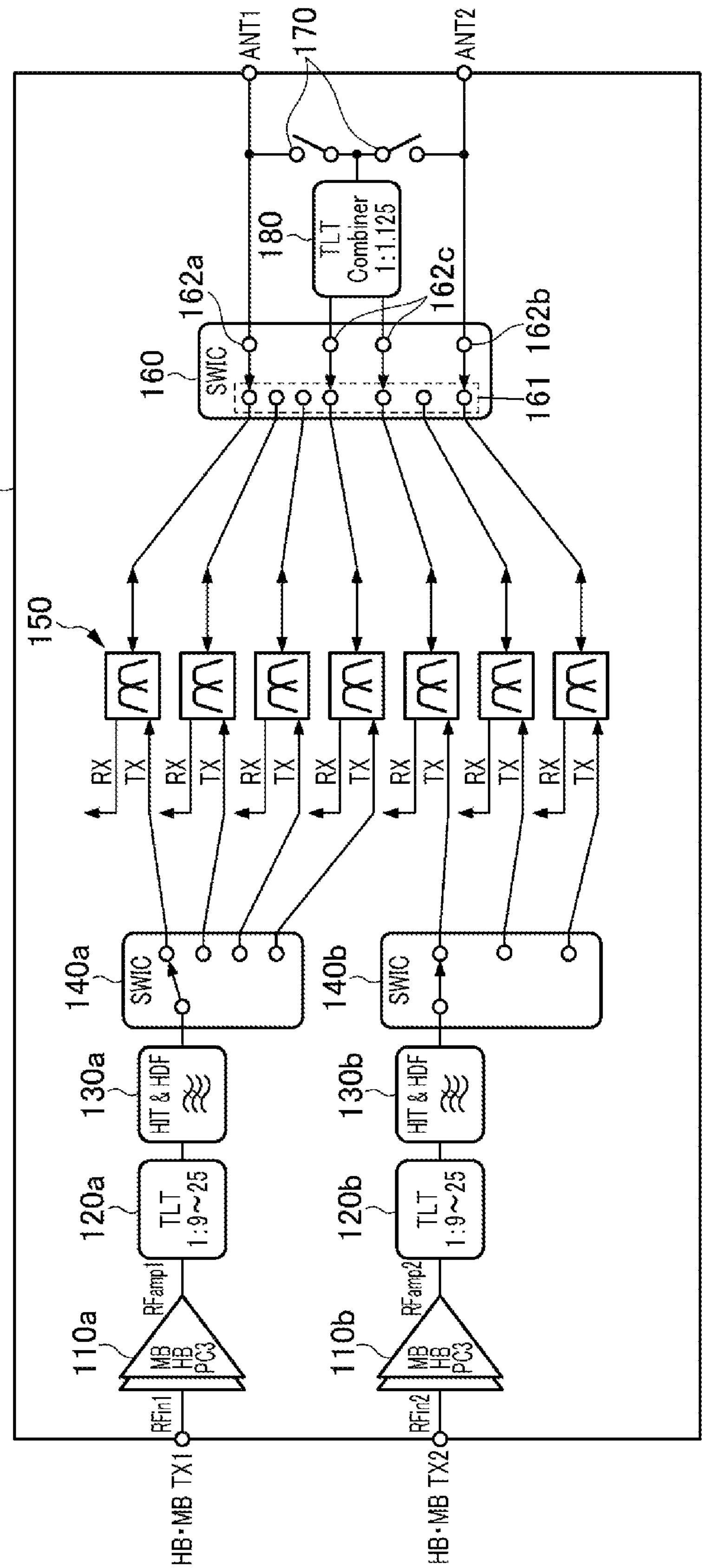
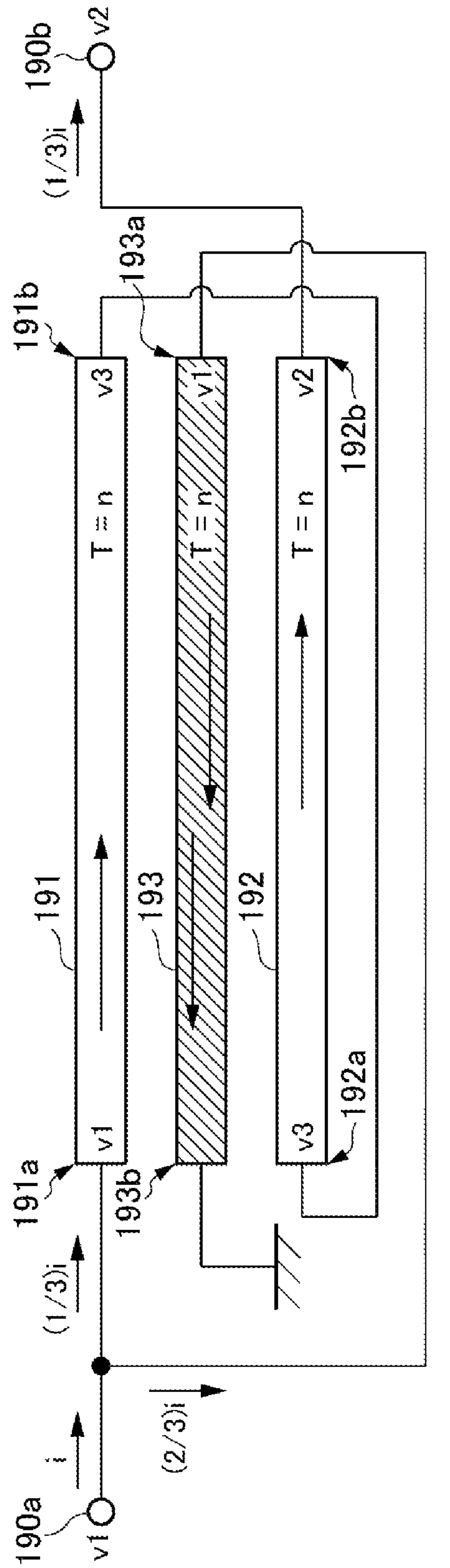

ANT2
170
180
TLT
Combiner
1:1.125
160
ANTSW
150
DPX
150
DPX
140a
BSSW
BSSW
140b
110a
MHB
PC3
MHB
PC3
110b
100
HB・MB TX1
HB・MB TX2
ANT1

FIG. 1

FIG. 2


ANT1
ANT2
170
180
TLT
Combiner
1:1.125
162a
162c
162b
160
SWIC
161
100
150
RX
TX
140a
SWIC
140b
SWIC
130a
HIT & HDF
130b
HIT & HDF
120a
TLT
1:9~25
120b
TLT
1:9~25
110a
MB
HB
PC3
110b
MB
HB
PC3
RFamp1
RFamp2
RFin1
RFin2
HB·MB TX1
HB·MB TX2



190b
v2
(1/3)i
193a
191b
193a
v3
v1
v2
T = n
T = n
T = n
192b
191
193
192
192a
v1
v3
191a
193b
(1/3)i
(2/3)i
i
190a
v1

FIG. 3A

FIG. 3B

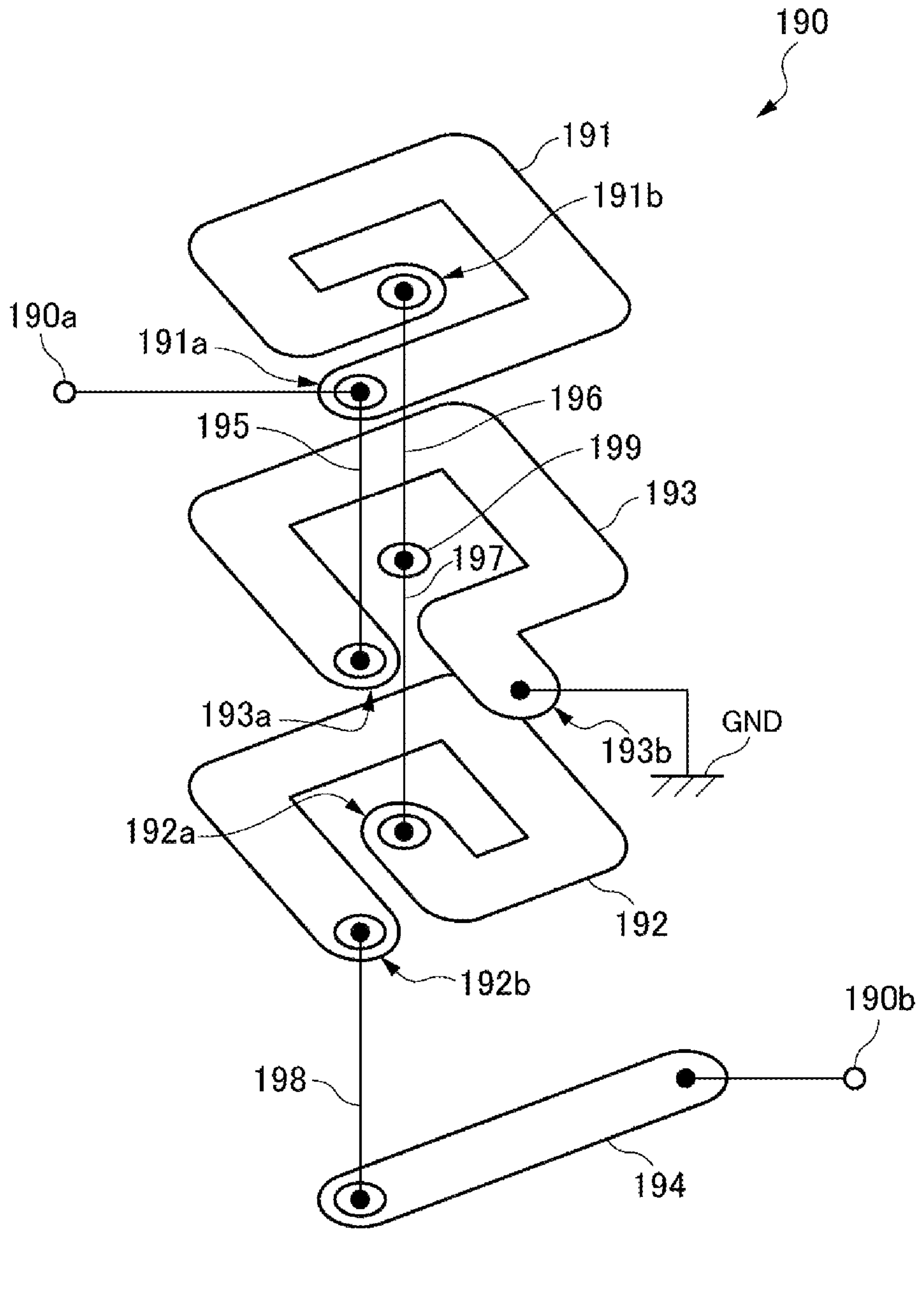
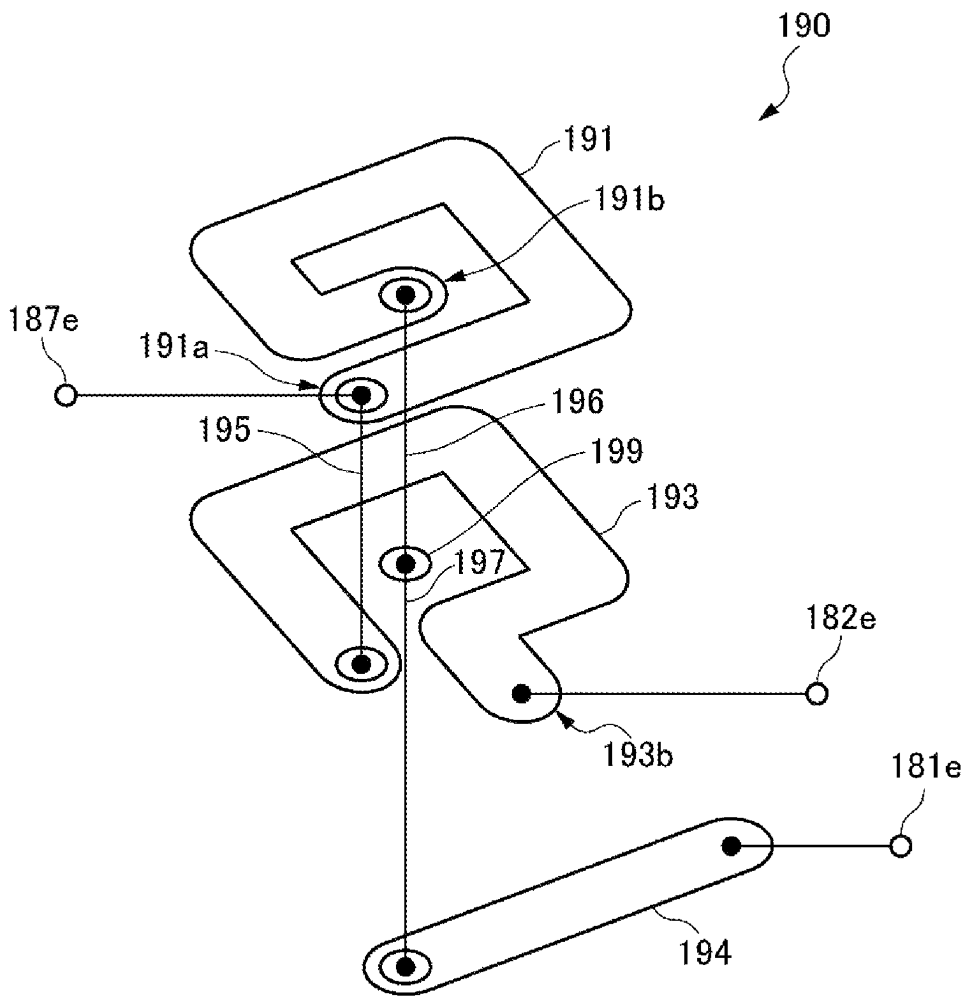

190
191
191b
190a
191a
196
195
199
193
197
193a
193b
GND
192a
192
192b
190b
198
194

FIG. 3C



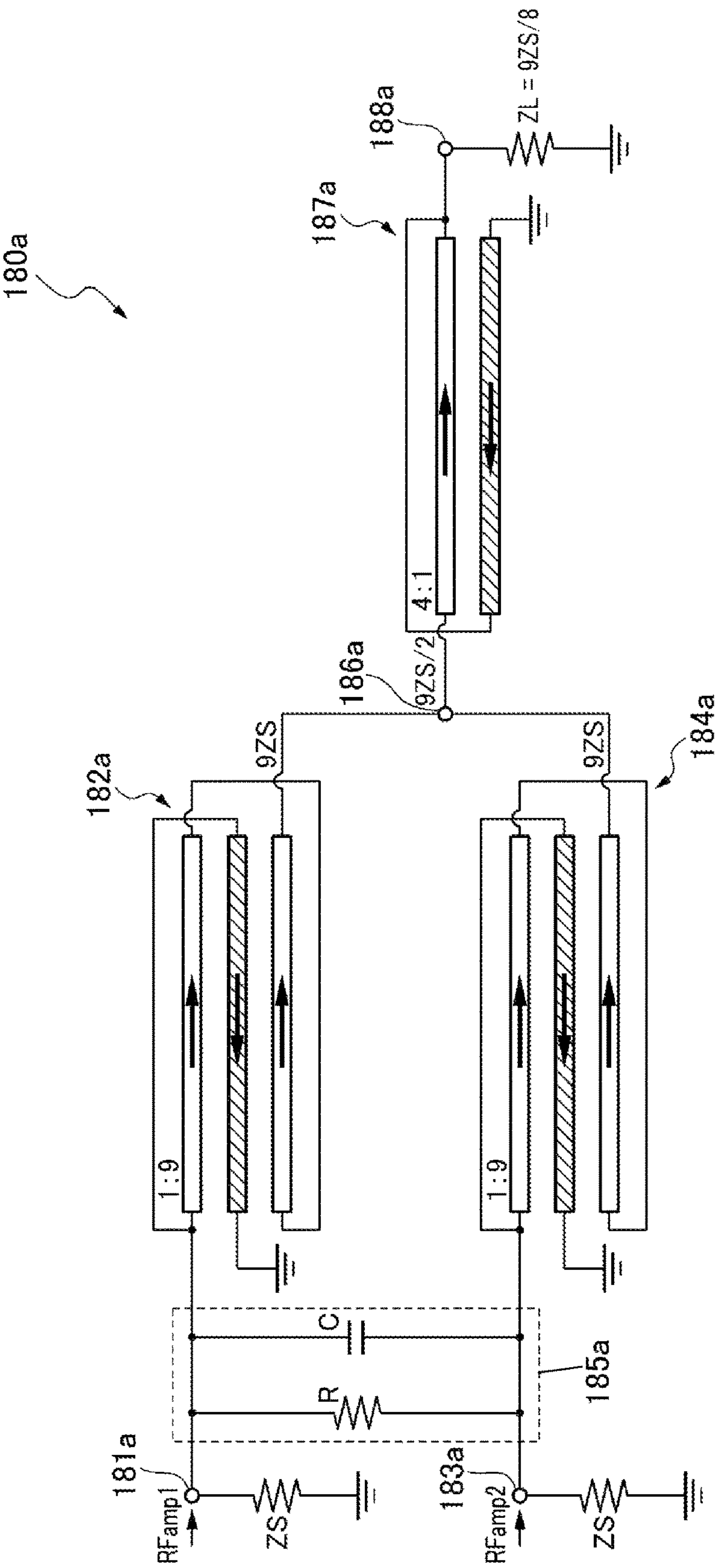

180a
188a
ZL = 9ZS/8
187a
4:1
186a
9ZS/2
182a
9ZS
9ZS
184a
1:9
1:9
C
R
185a
181a
183a
RFamp1
RFamp2
ZS
ZS

FIG. 4

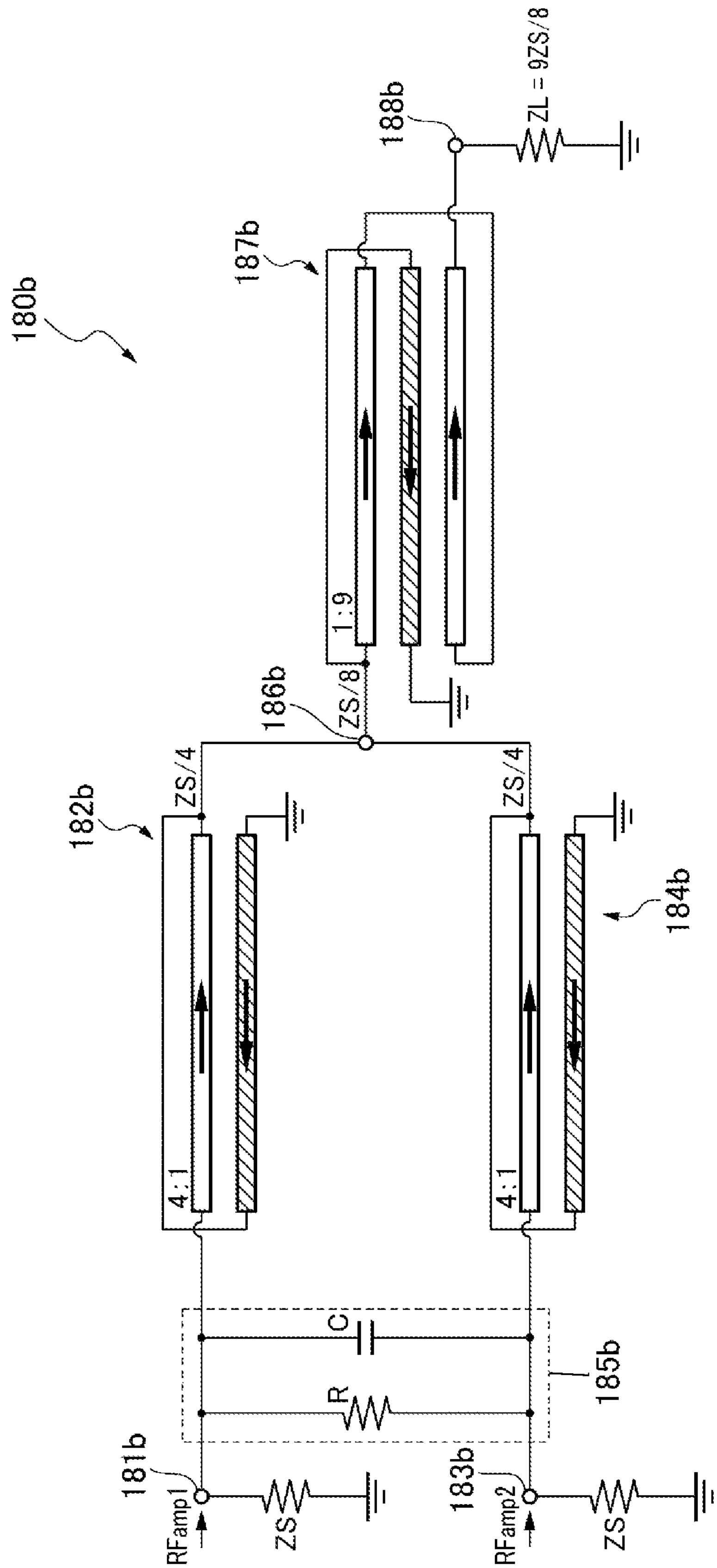
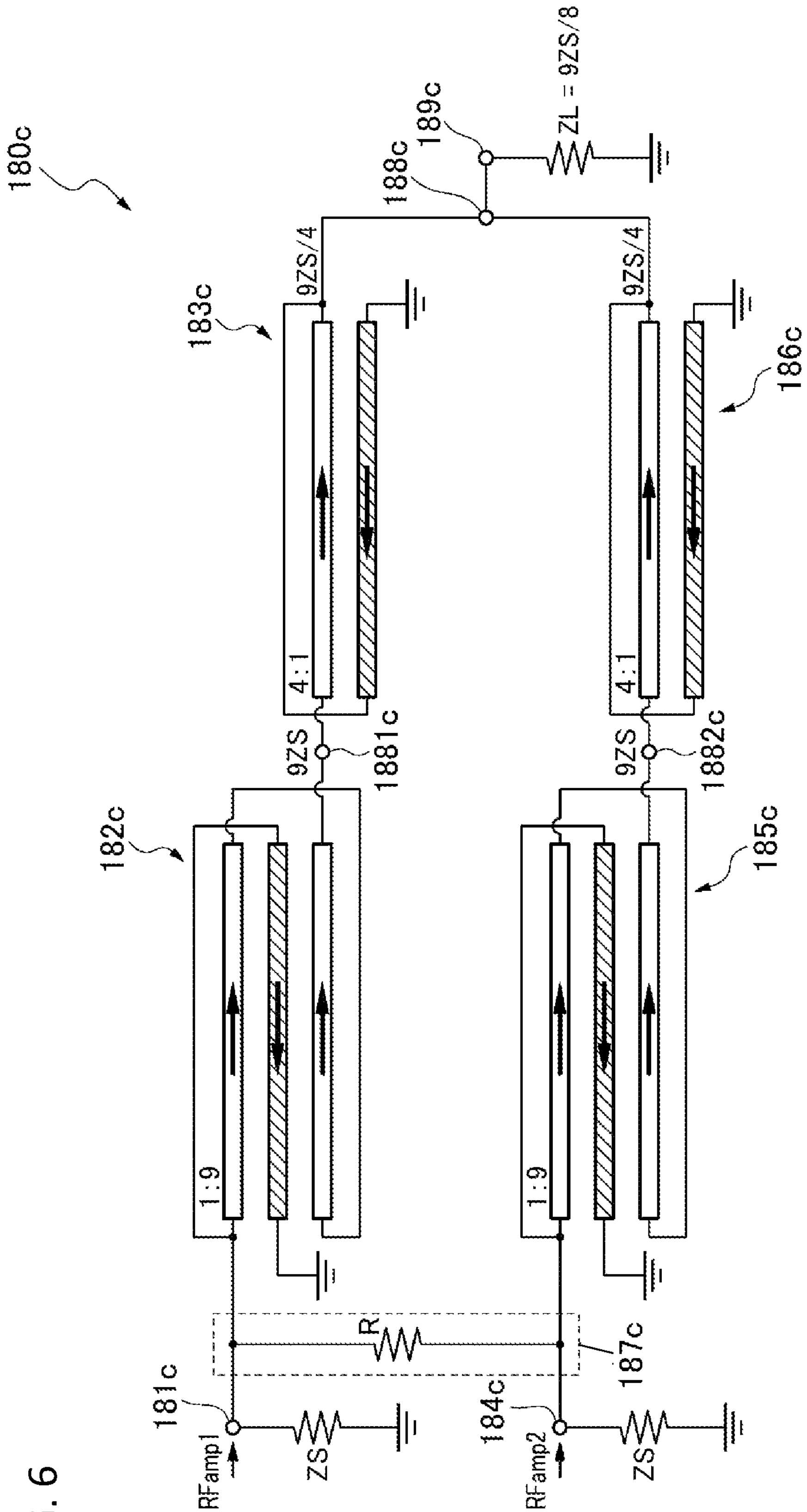

180b
187b
188b
ZL = 9ZS/8
1:9
186b
ZS/8
182b
ZS/4
ZS/4
184b
4:1
4:1
C
R
185b
181b
183b
RFamp1
RFamp2
ZS
ZS

FIG. 5

FIG. 6


180c
189c
ZL = 9ZS/8
188c
9ZS/4
9ZS/4
183c
186c
4:1
4:1
9ZS
1881c
9ZS
1882c
182c
185c
1:9
1:9
R
187c
181c
184c
RFamp1
RFamp2
ZS
ZS

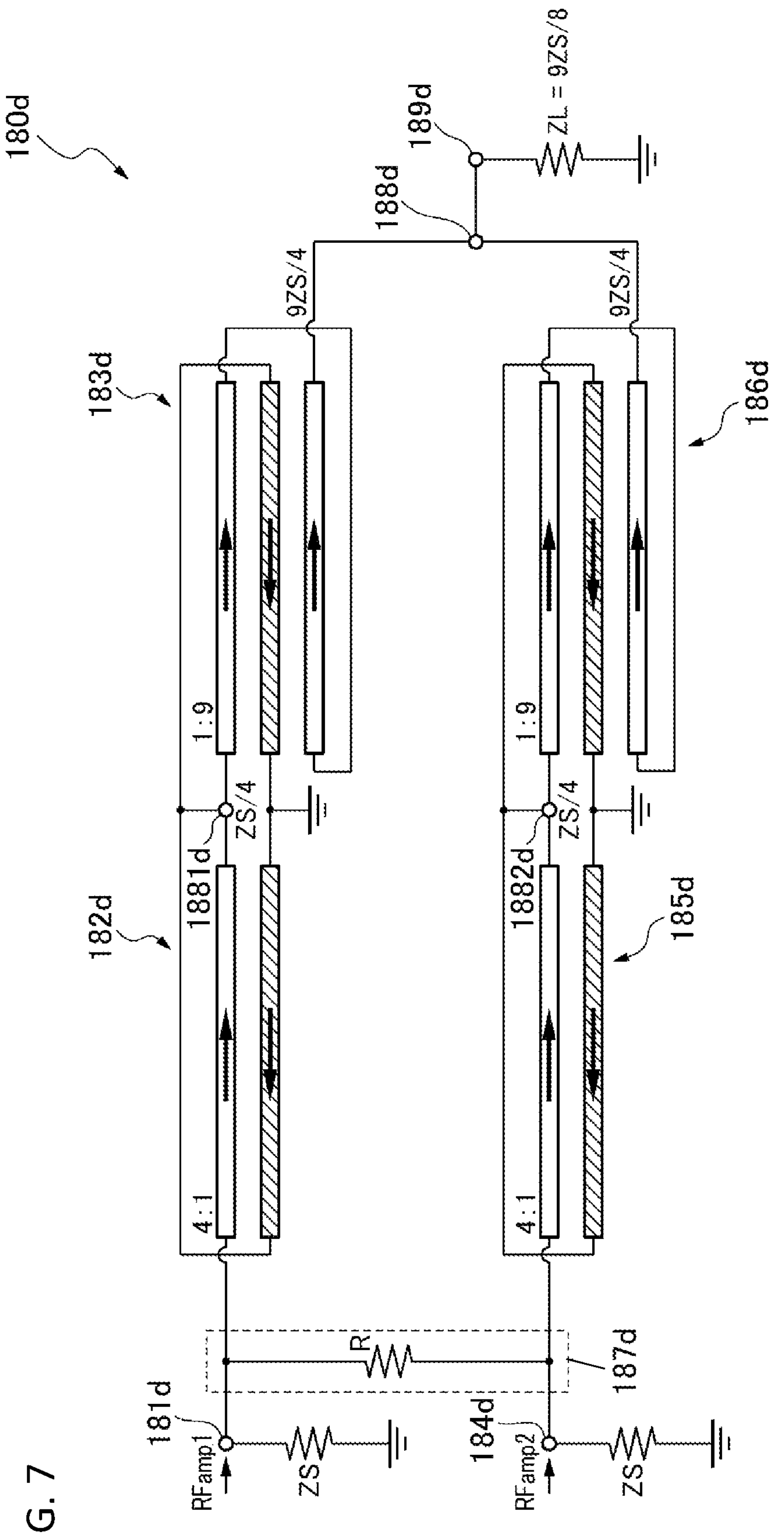

180d
189d
ZL = 9ZS/8
188d
9ZS/4
9ZS/4
183d
186d
1:9
1:9
ZS/4
ZS/4
1881d
1882d
182d
185d
4:1
4:1
R
187d
181d
184d
RFamp1
RFamp2
ZS
ZS

FIG. 7

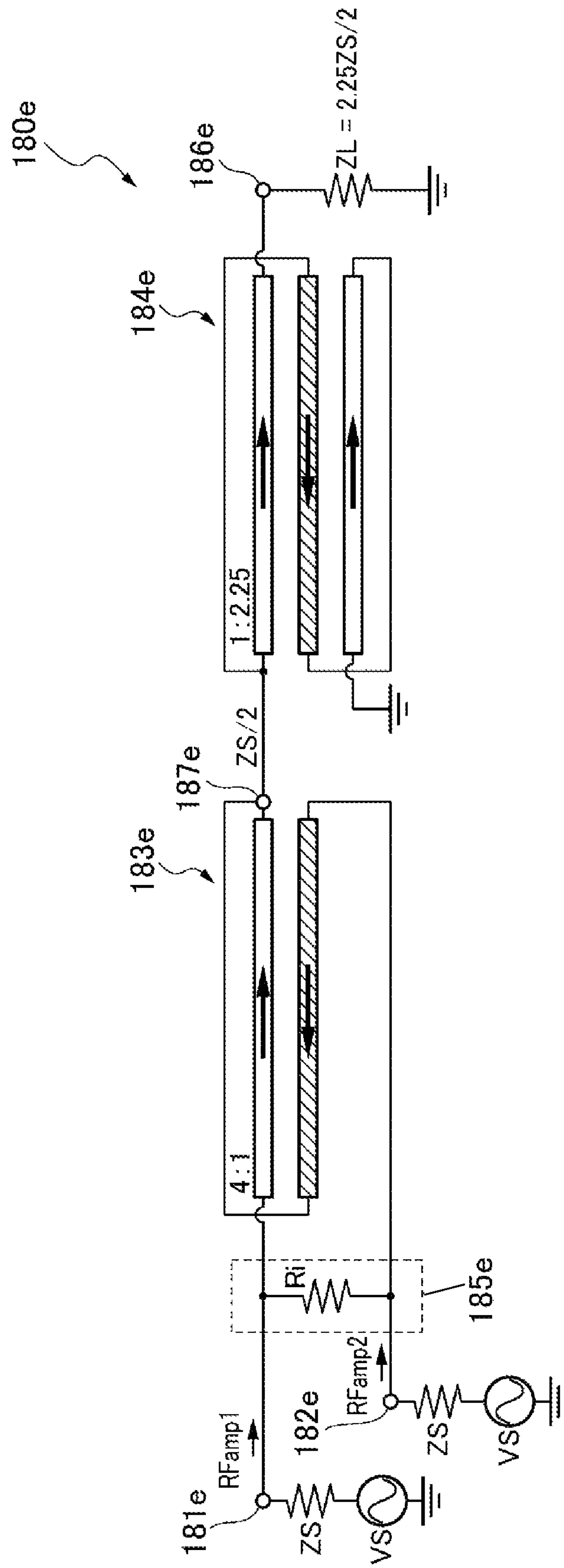

180e
186e
ZL = 2.25ZS/2
184e
1:2.25
ZS/2
187e
183e
4:1
Ri
185e
RFamp2
182e
ZS
VS
RFamp1
181e
ZS
VS

FIG. 8

FIG. 9A

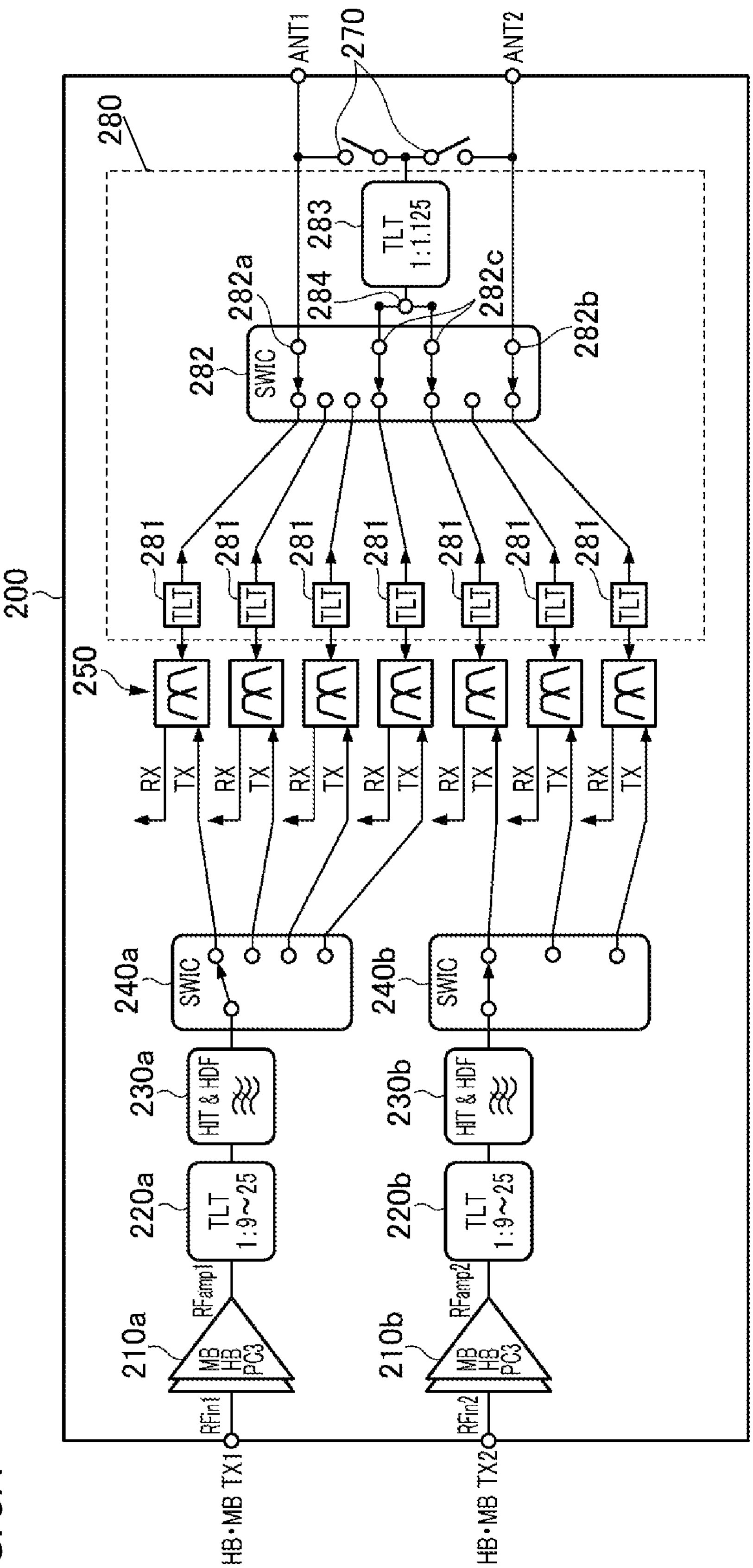
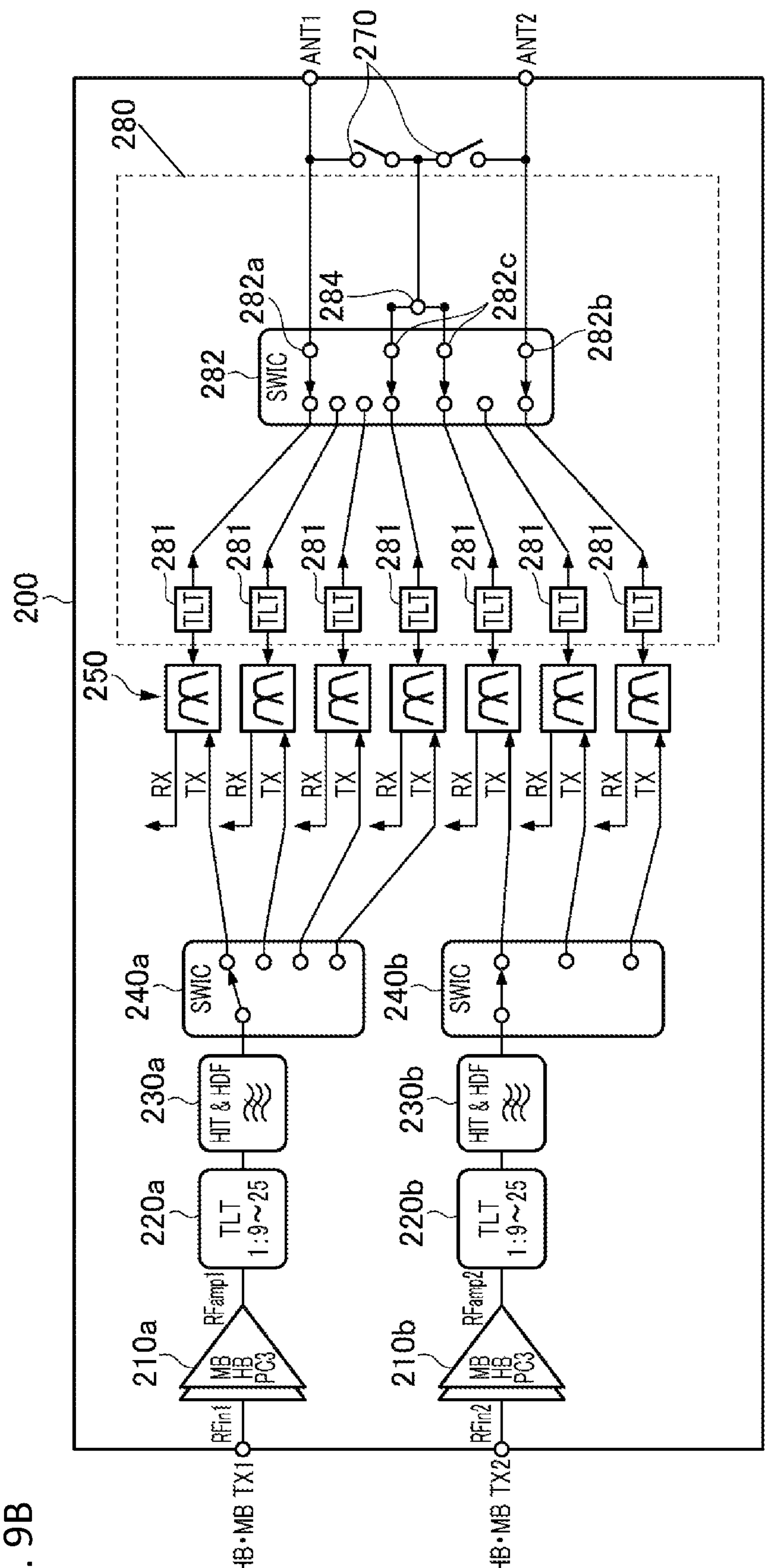

200
280
270
ANT1
ANT2
TLT 1:1.125
283
284
282a
282c
282b
282
SWIC
281
TLT
250
RX
TX
240a
SWIC
240b
230a
HIT & HDF
230b
220a
TLT 1:9~25
220b
210a
MB HB PC3
210b
RFamp1
RFamp2
RFin1
RFin2
HB·MB TX1
HB·MB TX2

FIG. 9B



ANT1
ANT2
270
280
200
284
282a
282b
282c
282
SWIC
281
TLT
250
RX
TX
240a
240b
SWIC
230a
230b
HIT & HDF
220a
220b
TLT
1:9~25
210a
210b
MB
HB
PC3
RFamp1
RFamp2
RFin1
RFin2
HB·MB TX1
HB·MB TX2

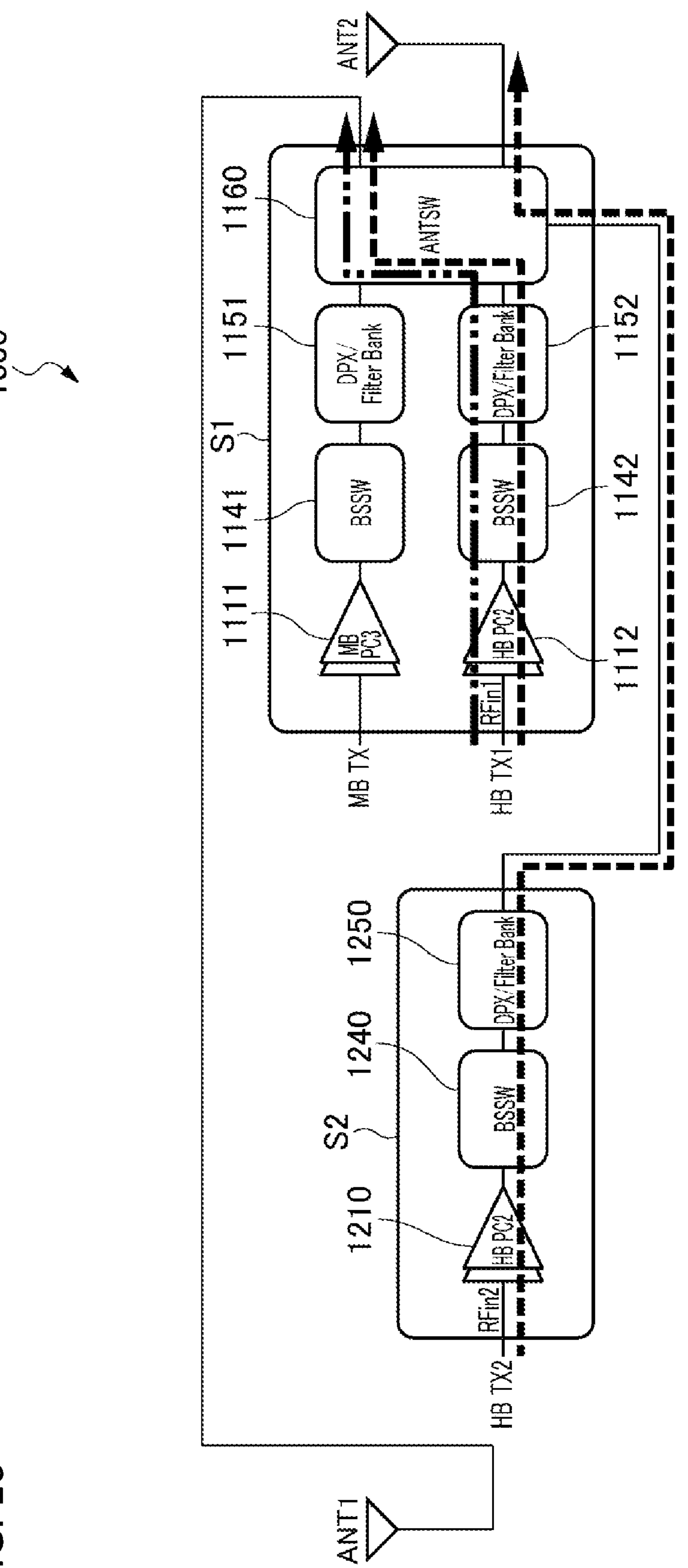

1000
ANT2
ANT1
1160
ANTSW
1151
DPX/
Filter Bank
1152
DPX/Filter Bank
S1
1141
BSSW
1142
BSSW
1111
MB
PC3
1112
HB PC2
RFin1
MB TX
HB TX1
1250
DPX/Filter Bank
1240
BSSW
S2
1210
HB PC2
RFin2
HB TX2

FIG. 10

POWER AMPLIFIER MODULE

CROSS REFERENCE TO RELATED APPLICATION

This application claims priority from Japanese Patent Application No. 2020-122668 filed on Jul. 17, 2020. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

The present disclosure relates to a power amplifier module.

2. Description of the Related Art

In recent years, in mobile terminals, for a radio frequency (RF) front-end circuit, a device supporting multiple frequency bands specified in the third generation partnership project (3GPP) is used. Furthermore, high-speed communications are demanded, and multiband operation using multiple frequency bands simultaneously is therefore adopted. Thus, a technique is disclosed in which signals amplified by respective amplifiers of two systems are combined to generate a signal compliant with an intended communication standard (see Japanese Unexamined Patent Application Publication No. 2016-42699).

In a front-end circuit disclosed in Japanese Unexamined Patent Application Publication No. 2016-42699, 3G/4G signals of the two systems are combined by a power combiner including a matching network and a band selection switch, and then a 2G signal is outputted through an impedance transformation circuit. However, in the front-end circuit, an LC ladder circuit is used for the matching network, and thus a wider bandwidth is not able to be achieved.

Furthermore, International Publication No. WO2015/133003A1 discloses an amplifier circuit that combines amplified signals. The amplifier circuit disclosed in International Publication No. WO2015/133003A1 combines signals through respective quarter-wave lines provided in two respective systems to generate an output signal. However, in the amplifier circuit disclosed in International Publication No. WO2015/133003A1, output matching is performed by using quarter-wave lines whose optimum line lengths differ according to a frequency, and thus it is difficult to achieve a wider bandwidth. In other words, in the amplifier circuit, wideband signals are not able to be combined. Furthermore, the area occupied by each quarter-wave line in the circuit is large, thus resulting in an increase in module size.

BRIEF SUMMARY OF THE DISCLOSURE

Thus, the present disclosure has been made to appropriately combine wideband signals while achieving a reduction in module size.

A power amplifier module according to one aspect of the present disclosure includes a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second

impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level and including a transmission line transformer.

The present disclosure can provide the power amplifier module that can appropriately combine wideband signals while achieving a reduction in module size. Other features, elements, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of preferred embodiments of the present disclosure with reference to the attached drawings.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. **1** illustrates an overview of a configuration of a power amplifier module according to a present embodiment;

FIG. **2** illustrates an example of the configuration of the power amplifier module according to the present embodiment;

FIG. **3**A schematically illustrates an example of a transmission line transformer;

FIG. **3**B is a perspective view schematically illustrating an example of a structure of the transmission line transformer;

FIG. **3**C is a perspective view schematically illustrating an example of the structure of the transmission line transformer;

FIG. **4** illustrates an example of a configuration of a first form of combiner;

FIG. **5** illustrates an example of a configuration of a second form of combiner;

FIG. **6** illustrates an example of a configuration of a third form of combiner;

FIG. **7** illustrates an example of a configuration of a fourth form of combiner;

FIG. **8** illustrates an example of a configuration of a fifth form of combiner;

FIG. **9**A illustrates an overview of a configuration of a power amplifier module according to another embodiment;

FIG. **9**B illustrates an overview of a configuration of the power amplifier module according to the other embodiment; and

FIG. **10** illustrates an overview of a configuration of a power amplifier module according to a comparative example.

DETAILED DESCRIPTION OF THE DISCLOSURE

Embodiments of the present disclosure will be described below with reference to the drawings. Here, circuit elements denoted by the same reference numerals are the same circuit element, and repeated descriptions of the circuit elements are omitted.

Configuration of Power Amplifier Module **100**

An overview of a power amplifier module **100** according to a present embodiment will be described with reference to FIG. **1**. FIG. **1** illustrates an overview of a configuration of the power amplifier module **100** according to the present embodiment. The power amplifier module **100** is incorporated, for example, in a mobile communication device, such as a cellular phone. The power amplifier module **100** amplifies the power of an input signal RFin to a level necessary for transmission to a base station and outputs this input signal RFin as an amplified signal RFout. The input signal RFin is a radio frequency (RF) signal modulated, for example, by a radio frequency integrated circuit (RFIC) or the like in accordance with a certain communication scheme. Examples of a communication standard of the input signal RFin include the second generation mobile communication system (2G), the third generation mobile communication system (3G), the fourth generation mobile communication system (4G), the fifth generation mobile communication system (5G), 5G new radio (5GNR), long term evolution (LTE)-frequency division duplex (FDD), LTE-time division duplex (TDD), LTE-Advanced, and LTE-Advanced Pro. The frequency of the input signal RFin ranges, for example, from about several hundred MHz to about several tens of GHz. The communication standard and frequency of the input signal RFin are not limited to these.

For example, between a mobile terminal using downlink multiple-input and multiple-output (MIMO), such as a mobile terminal supporting 5GNR, and a base station, a reference signal for grasping a radio wave propagation situation is transmitted before data communication. At this time, a reference signal is outputted to two antennas of a plurality of antennas incorporated in the mobile terminal. On the other hand, when an uplink is established from the mobile terminal to the base station, a PC2 transmission from one antenna and PC3 transmissions from two respective antennas are performed.

Here, each of PC2 and PC3 denotes "Power Class" representing an output level of a signal and is specified in 3GPP. With respect to PC3, for example, transmission power transmitted from an antenna of the mobile terminal is about 23 dBm. With respect to PC2, for example, transmission power transmitted from an antenna of the mobile terminal is about 26 dBm. In other words, PC2 is higher than PC3 in transmission power by about 3 dBm, and thus the output of PC2 is about twice the output of PC3.

Hence, in mobile terminals supporting 5GNR, for example, an aspect in which a reference signal is transmitted, an aspect in which a PC2 transmission from one antenna is performed, and an aspect in which PC3 transmissions from two respective antennas are performed have to be implemented. For at least these aspects, the power amplifier module **100** according to the present embodiment can achieve high-efficiency transmission.

First, an overview of the above-described aspects in the power amplifier module **100** will be described with reference to FIG. **1**. In FIG. **1**, as an example, communication paths that transmit PC3 signals from two respective antennas ANT**1** and ANT**2** are indicated by a dashed line, and a communication path that transmits a PC2 signal from one antenna ANT**1** (or ANT**2**) is indicated by a dash-dot-dot line.

As illustrated in FIG. **1**, an input signal RFin**1** is inputted from a terminal TX**1** to a first amplifier **110***a*, and an input signal RFin**2** is inputted from a terminal TX**2** to a second amplifier **110***b*. The first amplifier **110***a* and the second amplifier **110***b* are optimally designed, for example, for PC3.

In the power amplifier module **100**, when PC3 signals are transmitted from the two respective antennas ANT**1** and ANT**2**, an input signal RFin**1** is transmitted to the antenna ANT**1** through the first amplifier **110***a*, a first band selection switch **140***a*, a duplexer **150**, and an antenna switch **160**. Similarly, an input signal RFin**2** is transmitted to the antenna ANT**2** through the second amplifier **110***b*, a second band selection switch **140***b*, a duplexer **150**, and the antenna switch **160**. Although not illustrated in FIG. **1**, behind the first and second band selection switches **140***a* and **140***b*, a plurality of duplexers **150** suited to respective bands may be connected.

On the other hand, in the power amplifier module **100**, when a PC2 signal is transmitted from one antenna ANT**2**, an input signal RFin**1** is inputted to a combiner **180** through the first amplifier **110***a*, the first band selection switch **140***a*, the duplexer **150**, and the antenna switch **160**. An input signal RFin**2** is inputted to the combiner **180** through the second amplifier **110***b*, the second band selection switch **140***b*, the duplexer **150**, and the antenna switch **160**. The combiner **180** combines the input signal RFin**1** of PC3 (amplified signal RFamp**1**) and the input signal RFin**2** of PC3 (amplified signal RFamp**2**) and outputs a PC2 signal. Here, the combiner **180** includes a transmission line transformer. The signal outputted from the combiner **180** is transmitted to the antenna ANT**2** through a path-changing switch **170**.

In other words, the power amplifier module **100** can transmit PC3 signals to the respective antennas ANT**1** and ANT**2** of two systems by using two amplifiers optimally designed for PC3 and can further transmit a PC2 signal to the antenna ANT**1** (or ANT**2**) of one system.

Here, a power amplifier module **1000** according to a comparative example will be described with reference to FIG. **10**. FIG. **10** illustrates an overview of a configuration of the power amplifier module **1000** according to the comparative example. In FIG. **10**, as an example, communication paths that transmit PC3 signals from the two respective antennas ANT**1** and ANT**2** are indicated by a dashed line, and a communication path that transmits a PC2 signal from one antenna ANT**1** is indicated by a dash-dot-dot line.

In the power amplifier module **1000**, when PC3 signals are transmitted from the two respective antennas ANT**1** and ANT**2**, an input signal RFin**1** is inputted to an amplifier **1112** provided in or on a given substrate S**1**, and an input signal RFin**2** is inputted to an amplifier **1210** provided in or on a substrate S**2** different from the given substrate S**1**. Here, each of the amplifiers **1112** and **1210** is optimally designed for PC2 to implement a PC2 transmission using one antenna. The input signals RFin**1** and RFin**2** are transmitted to the respective antennas ANT**1** and ANT**2** through band selection switches **1142** and **1240** and duplexers **1152** and **1250** that are provided in or on the different respective substrates S**1** and S**2**, and through an antenna switch **1160** provided in or on the given substrate S**1**.

As described above, in comparison with the power amplifier module **1000** according to the comparative example, in the power amplifier module **100** according to the present embodiment, for example, the first amplifier **110***a* and the second amplifier **110***b* are optimally designed for PC3, thus enabling an increase in transmission efficiency.

Furthermore, in the power amplifier module **100**, for example, PC3 signals are combined to generate a PC2 signal, and a component (substrate S**2** of the power amplifier module **1000**) for generating a PC2 signal is therefore unnecessary, enabling a reduction in the number of components and a reduction in package size. Additionally, since the combiner **180** includes the transmission line transformer, a wide frequency band including a middle band (MB) and a high band (HB) can be supported by one combiner **180**. Furthermore, the combiner **180** is disposed on an antennas ANT**1** and ANT**2** side of the antenna switch **160**, and thus impedances at input and output terminals of the antenna switch **160** are about 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch **160**.

Next, the configuration of the power amplifier module **100** will be described in detail with reference to FIG. **2**. FIG. **2** illustrates an example of the configuration of the power amplifier module **100** according to the present embodiment.

The power amplifier module **100** includes, for example, the first amplifier **110***a*, a first impedance transformer **120***a*, a first filter circuit **130***a*, the first band selection switch **140***a*, the second amplifier **110***b*, a second impedance transformer **120***b*, a second filter circuit **130***b*, the second band selection switch **140***b*, duplexers **150**, the antenna switch **160**, the path-changing switch **170**, and the combiner **180**. Each component will be described in detail below.

The first amplifier **110***a* is, for example, a circuit that amplifies a power level of an input signal RFin**1** and outputs an amplified signal RFamp**1**. The first amplifier **110***a* supports, for example, input signals RFin**1** in a frequency band (about 1710 MHz to about 2690 MHz) including the MB and the HB. The first amplifier **110***a* is optimally designed, for example, for PC3.

The first impedance transformer **120***a* is, for example, a circuit that performs impedance transformation at a predetermined transformation ratio, and includes a transmission line transformer. In FIG. **2**, as an example, a transformation ratio of "1:9 to 25" is indicated. When the transmission line transformer is included, wideband and low-loss impedance transformation can be achieved. A transmission line transformer will be described in detail below.

The first filter circuit **130***a* subjects, for example, a signal in the frequency band including the MB and the HB to filtering. The first filter circuit **130***a* may be, for example, a fixed filter whose pass band is fixed at a specific frequency band, a variable filter whose pass band is varied in accordance with each of multiple frequency bands, or a filter obtained by combining the fixed filter and the variable filter such that they are switchable. Thus, attenuation characteristics in proximity to the pass band can be increased.

The first band selection switch (BSSW) **140***a* is, for example, a switch that distributes a plurality of high-frequency signals whose frequency bands are different to intended terminals. In the first band selection switch **140***a*, the amplified signal RFamp**1** is inputted to an input terminal, and this signal is outputted from an intended output terminal.

The second amplifier **110***b*, the second impedance transformer **120***b*, the second filter circuit **130***b*, and the second band selection switch **140***b* are similar to the first amplifier **110***a*, the first impedance transformer **120***a*, the first filter circuit **130***a*, and the first band selection switch **140***a*, and thus a description thereof is omitted.

In other words, high-frequency signals being substantially equal in phase and amplitude are inputted to the first amplifier **110***a* and the second amplifier **110***b*. Then, the two amplifiers **110***a* and **110***b* amplify the input high-frequency signals and output the high-frequency signals being substantially equal in phase and amplitude. The high-frequency signals amplified by the two amplifiers **110***a* and **110***b* are subjected to impedance transformation in transmission line transformers in the respective impedance transformers **120***a* and **120***b* and then are outputted to duplexers **150**.

Each of the duplexers (DPXs) **150** is, for example, a filter circuit that, when a transmission frequency and a reception frequency are different, switches between transmission and reception. The duplexers **150** are connected, for example, between output terminals of the first and second band selection switches **140***a* and **140***b* and input terminals **161** of the antenna switch **160** to be described. FIG. **2** illustrates distribution of an amplified signal RFamp**1** and a reception signal RX by a duplexer **150**. Incidentally, if a communication scheme is, for example, a time division duplex (TDD) scheme, the duplexers **150** may be omitted. Instead, the duplexer **150** may be replaced by typical band pass filters.

The antenna switch (ANTSW) **160** is, for example, a switch that switches between a path connected to each of the antennas ANT**1** and ANT**2** and a path connected to the combiner **180** to be described. Thus, it can be determined selectively whether to output PC3 signals to the respective antennas ANT**1** and ANT**2** or to input two PC3 signals to the combiner **180** to output a PC2 signal.

Specifically, the antenna switch **160** includes, for example, the input terminals **161**, a first output terminal **162***a*, a second output terminal **162***b*, and third output terminals **162***c*. The input terminals **161** are, for example, terminals that are connected to the respective duplexers **150** and to which an amplified signal RFamp**1** and an amplified signal RFamp**2** are inputted. The first output terminal **162***a* is, for example, a terminal connected to the antenna ANT**1** so as to output the amplified signal RFamp**1** to the antenna ANT**1**. The second output terminal **162***b* is, for example, a terminal connected to the antenna ANT**2** so as to output the amplified signal RFamp**2** to the antenna ANT**2**. The third output terminals **162***c* are, for example, terminals connected to the combiner **180** so as to output the amplified signal RFamp**1** and the amplified signal RFamp**2** to the combiner **180**.

The path-changing switch **170** is a switch that distributes a signal outputted from the combiner **180** to be described to an intended antenna.

The combiner **180** is, for example, a circuit that combines PC3 signals that are the amplified signals RFamp**1** and RFamp**2** to generate a PC2 signal. The combiner **180** is connected between the antenna switch **160** and the antennas ANT**1** and ANT**2**. Thus, impedances at all input and output terminals of the antenna switch **160** are substantially 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch **160**. The combiner **180** is constructed, for example, by combining transmission line transformers. When the combiner **180** includes the transmission line transformers, wideband and low-loss impedance transformation can be achieved.

First, a structure of a transmission line transformer **190** will be described below with reference to FIGS. **3**A and **3**B. Then, variations of the combiner **180** obtained by combining transmission line transformers **190** will be described with reference to FIGS. **4** to **8**.

FIG. **3**A schematically illustrates an example of the transmission line transformer **190**. FIGS. **3**B and **3**C are perspective views schematically illustrating an example of the structure of the transmission line transformer **190**.

As illustrated in FIG. **3**A, the transmission line transformer **190** includes, for example, a first transmission line **191**, a second transmission line **192**, and a third transmission line **193**. First, alternating currents (AC) that flow through the first transmission line **191**, the second transmission line **192**, and the third transmission line **193** will be described. A current that flows from an input terminal **190***a* toward an output terminal **190***b* flows first from, of the first transmission line **191**, a first end portion **191***a* toward a second end portion **191***b*. Then, the current flows from a third end portion **192***a* toward a fourth end portion **192***b* in the second transmission line **192**. The magnitude of the AC current that flows through the first transmission line **191** is equal to the magnitude of the AC current that flows through the second transmission line **192**. In the third transmission line **193**, an odd mode current is induced from a fifth end portion **193***a* toward a sixth end portion **193***b* by the AC current that flows through the first transmission line **191**. In the third transmission line **193**, an odd mode current is induced from the fifth end portion **193***a* toward the sixth end portion **193***b* by the AC current that flows through the second transmission line **192**. A direction of an odd mode current induced in the third transmission line **193** is opposite to directions of the AC currents that flow through the first transmission line **191** and the second transmission line **192**. The odd mode current caused by the current that flows through the first transmission line **191** and the odd mode current caused by the current that flows through the second transmission line **192** are equal in magnitude and direction.

In other words, in the third transmission line **193**, the odd mode current derived from the first transmission line **191** and the odd mode current derived from the second transmission line **192** are superimposed and flow. For this reason, in the third transmission line **193**, an odd mode current is induced whose magnitude is twice the magnitude of a current that flows through a series circuit composed of the first transmission line **191** and the second transmission line **192**. When the magnitude of a current that flows from the input terminal **190***a* into the transmission line transformer **190** is denoted by i, a current of (⅓)i flows through the series circuit composed of the first transmission line **191** and the second transmission line **192**, and a current of (⅔)i flows through the third transmission line **193**. Hence, the magnitude of the current outputted from the output terminal **190***b* is (⅓)i.

Next, a voltage will be described. A voltage at the input terminal **190***a* is denoted by v1, and a voltage at the output terminal **190***b* is denoted by v2. A voltage at the first end portion **191***a* of the first transmission line **191** and a voltage at the fifth end portion **193***a* of the third transmission line **193** are equal to the voltage v1 at the input terminal **190***a*. A voltage at the fourth end portion **192***b* of the second transmission line **192** is equal to the voltage v2 at the output terminal **190***b*. A voltage at the second end portion **191***b* of the first transmission line **191** is denoted by v3. A voltage at the third end portion **192***a* of the second transmission line **192** is equal to the voltage v3 at the second end portion **191***b* of the first transmission line **191**. A voltage at the sixth end portion **193***b* of the third transmission line **193** is 0 V.

A potential difference between the first end portion **191***a* and the second end portion **191***b* of the first transmission line **191** is equal to a potential difference between the sixth end portion **193***b* and the fifth end portion **193***a* of the third transmission line **193**, and thus an equation of $v1-v3=0-v1$ holds. Similarly, for a relationship between the second transmission line **192** and the third transmission line **193**, an equation of $v3-v2=0-v1$ holds. When the simultaneous equations are solved, $3\times v1=v2$ is obtained. Thus, the voltage v2 at the output terminal **190***b* is three times the voltage v1 at the input terminal **190***a*.

When a load having an impedance R2 is connected to the output terminal **190***b*, an equation of $v2=(⅓)i\times R2$ holds. When an impedance as seen from the input terminal **190***a* looking toward a load side is denoted by R1, an equation $v1=R1\times i$ holds. When these equations are solved, $R1=(1/9)R2$ is obtained. Thus, the impedance R1 as seen from the input terminal **190***a* looking toward the load side is 1/9 times the impedance R2 of the load connected to the output terminal **190***b*. On the other hand, when the load is connected to the input terminal **190***a*, an impedance as seen from the output terminal **190***b* looking toward a load side is nine times an impedance of the load connected to the input terminal **190***a*. Thus, the transmission line transformer **190** according to the present embodiment functions as an impedance transformation circuit with an impedance transformation ratio of nine.

As illustrated in FIG. **3**B, in the transmission line transformer, the first to third transmission lines are disposed so as to be stacked in a thickness direction of a substrate. Specifically, the first end portion **191***a* of the first transmission line **191**, the fifth end portion **193***a* of the third transmission line **193**, and the fourth end portion **192***b* of the second transmission line **192** are disposed such that they overlay one another when viewed in plan. The second end portion **191***b* of the first transmission line **191** and the third end portion **192***a* of the second transmission line **192** are disposed such that they overlay each other when viewed in plan. In the same layer where the third transmission line **193** is disposed, a conductor pattern **199** is disposed at a position corresponding to the second end portion **191***b* of the first transmission line **191**. A via conductor **195** connects the first end portion **191***a* of the first transmission line **191** and the fifth end portion **193***a* of the third transmission line **193**. A via conductor **196** connects the second end portion **191***b* of the first transmission line **191** and the conductor pattern **199**, and a via conductor **197** connects the conductor pattern **199** and the third end portion **192***a* of the second transmission line **192**. A via conductor **198** connects the fourth end portion **192***b* of the second transmission line **192** and an extended line **194**. The first end portion **191***a* of the first transmission line **191** is connected to the input terminal **190***a*, and the extended line **194** is connected to the output terminal **190***b*. The sixth end portion **193***b* of the third transmission line **193** is connected to a ground conductor GND.

As described above, the transmission line transformer **190** illustrated in FIGS. **3**A and **3**B has an impedance transformation ratio of 1:9, whereas, as illustrated in FIG. **3**C, the transmission line transformer **190**, for example, with a two-layer structure composed of the first transmission line **191** and the third transmission line **193** has an impedance transformation ratio of 1:4. Furthermore, in the transmission line transformer **190** illustrated in FIGS. **3**A and **3**B, an impedance transformation ratio of 1:2.25 is provided by connecting the first end portion **191***a* of the first transmission line **191** to the fifth end portion **193***a* of the third transmission line **193**, connecting the sixth end portion **193***b* of the third transmission line **193** to the fourth end portion **192***b* of the second transmission line **192**, and connecting the third end portion **192***a* of the second transmission line **192** to the ground conductor GND.

In the present embodiment, there will be described, as an example, the combiner **180** constructed by any combination of a transmission line transformer with an impedance transformation ratio of 1:9 (or 9:1), a transmission line transformer with a transformation ratio of 1:4 (or 4:1), and a transmission line transformer with a transformation ratio of 1:2.25 (or 2.25:1). Note that the combiner **180** is constructed by combining transmission line transformers with various transformation ratios. For example, the combiner **180** only has to be constructed so that an impedance at an input terminal is substantially equal to an impedance at an output terminal. For convenience of explanation, the following description will be given assuming that a portion corresponding to an end portion connected to the input terminal **190***a* in the transmission line transformer **190** is referred to as “one end” and a portion corresponding to an end portion connected to the output terminal **190***b* is referred to as “the other end”.

First Form of Combiner **180***a*

A first form of combiner **180***a* will be described with reference to FIG. **4**. FIG. **4** illustrates an example of a configuration of the first form of combiner **180***a*.

As illustrated in FIG. **4**, the combiner **180***a* includes, for example, a first input terminal **181***a*, a first transmission line transformer **182***a*, a second input terminal **183***a*, a second transmission line transformer **184***a*, a first isolation section **185***a*, a combination terminal **186***a*, a third transmission line transformer **187***a*, and an output terminal **188***a*.

The first input terminal **181***a* is a terminal to which an amplified signal RFamp**1** is inputted. The first transmission line transformer **182***a* is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:9. In the first transmission line transformer **182***a*, one end is connected to the first input terminal **181***a*, and the other end is connected to the combination terminal **186***a*. The second input terminal **183***a* is a terminal to which an amplified signal RFamp**2** is inputted. The second transmission line transformer **184***a* is, for example, a circuit that performs impedance transformation at the same transformation ratio of 1:9 as that of the first transmission line transformer **182***a*. In the second transmission line transformer **184***a*, one end is connected to the second input terminal **183***a*, and the other end is connected to the combination terminal **186***a*. In other words, the first transmission line transformer **182***a* and the second transmission line transformer **184***a* are connected in parallel. The first isolation section **185***a* includes a resistor R and a capacitor C connected in parallel. The first isolation section **185***a* is connected between the first input terminal **181***a* and the second input terminal **183***a*. The combination terminal **186***a* is a terminal that combines the amplified signal RFamp**1** inputted through the first transmission line transformer **182***a* and the amplified signal RFamp**2** inputted through the second transmission line transformer **184***a*. The third transmission line transformer **187***a* is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the third transmission line transformer **187***a*, one end is connected to the combination terminal **186***a*, and the other end is connected to the output terminal **188***a*. The output terminal **188***a* is connected to the path-changing switch **170**.

Next, an overview of impedance transformation performed in the combiner **180***a* will be described. An impedance at the first input terminal **181***a* of the combiner **180***a* is ZS (50 ohms). In the first transmission line transformer **182***a*, ZS (50 ohms) is transformed into 9×ZS (450 ohms). The same things hold true for the second input terminal **183***a* and the second transmission line transformer **184***a*. Then, at the combination terminal **186***a*, an impedance of 9×ZS×½ (225 ohms) is reached and is transformed into a quarter in the third transmission line transformer **187***a*. Hence, an impedance (ZL) at the output terminal **188***a* is ZS×9/8 (about 50 ohms). Incidentally, in the third transmission line transformer **187***a*, the impedance may be finely adjusted.

Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

Second Form of Combiner **180***b*

A second form of combiner **180***b* will be described with reference to FIG. **5**. FIG. **5** illustrates an example of a configuration of the second form of combiner **180***b*.

As illustrated in FIG. **5**, the combiner **180***b* includes, for example, a third input terminal **181***b*, a fourth transmission line transformer **182***b*, a fourth input terminal **183***b*, a fifth transmission line transformer **184***b*, a second isolation section **185***b*, a combination terminal **186***b*, a sixth transmission line transformer **187***b*, and an output terminal **188***b*.

The second form of combiner **180***b* is a combiner in which, in the first form of combiner **180***a*, the first and second transmission line transformers **182***a* and **184***a* with the transformation ratio of 1:9 are changed to the fourth and fifth transmission line transformers **182***b* and **184***b* with a transformation ratio of 4:1 and the third transmission line transformer **187***a* with the transformation ratio of 4:1 is changed to the sixth transmission line transformer **187***b* with a transformation ratio of 1:9. This enables low-loss and wideband power combination.

Third Form of Combiner **180***c*

A third form of combiner **180***c* will be described with reference to FIG. **6**. FIG. **6** illustrates an example of a configuration of the third form of combiner **180***c*.

As illustrated in FIG. **6**, the combiner **180***c* includes, for example, a fifth input terminal **181***c*, a seventh transmission line transformer **182***c*, an eighth transmission line transformer **183***c*, a sixth input terminal **184***c*, a ninth transmission line transformer **185***c*, a tenth transmission line transformer **186***c*, a third isolation section **187***c*, a combination terminal **188***c*, a first connection terminal **1881***c*, a second connection terminal **1882***c*, and an output terminal **189***c*.

The fifth input terminal **181***c* is a terminal to which an amplified signal RFamp**1** is inputted. The seventh transmission line transformer **182***c* is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:9. In the seventh transmission line transformer **182***c*, one end is connected to the fifth input terminal **181***c*, and the other end is connected to one end of the eighth transmission line transformer **183***c* through the first connection terminal **1881***c*. The eighth transmission line transformer **183***c* is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the eighth transmission line transformer **183***c*, the other end is connected to the combination terminal **188***c*. The sixth input terminal **184***c* is a terminal to which an amplified signal RFamp**2** is inputted. The ninth transmission line transformer **185***c* is, for example, a circuit that performs impedance transformation at the same transformation ratio of 1:9 as that of the seventh transmission line transformer **182***c*. In the ninth transmission line transformer **185***c*, one end is connected to the sixth input terminal **184***c*, and the other end is connected to one end of the tenth transmission line transformer **186***c* through the second connection terminal **1882***c*. The tenth transmission line transformer **186***c* is, for example, a circuit that performs impedance transformation at the same transformation ratio of 4:1 as that of the eighth transmission line transformer **183***c*. In the tenth transmission line transformer **186***c*, the other end is connected to the combination terminal **188***c*. In other words, the seventh transmission line transformer **182***c* and the eighth transmission line transformer **183***c* are connected in parallel with the ninth transmission line transformer **185***c* and the tenth transmission line transformer **186***c*. The third isolation section **187***c* is connected between the fifth input terminal **181***c* and the sixth input terminal **184***c*. The third isolation section **187***c* is, for example, a resistor having a resistance value based on a power distribution ratio between a transmission line connected to the fifth input terminal **181***c* and a transmission line connected to the sixth input terminal **184***c*. The combination terminal **188***c* is a terminal that combines the amplified signal RFamp**1** inputted through the seventh transmission line transformer **182***c* and the eighth transmission line transformer **183***c* and the amplified signal RFamp**2** inputted through the ninth transmission line transformer **185***c* and the tenth transmission line transformer **186***c*. The output terminal **189***c* is connected to the combination terminal **188***c*. The output terminal **189***c* is connected to the path-changing switch **170**.

Next, an overview of impedance transformation performed in the combiner **180***c* will be described. An impedance at the fifth input terminal **181***c* of the combiner **180***c* is ZS (50 ohms). In the seventh transmission line transformer **182***c*, ZS (50 ohms) is transformed into 9×ZS (450 ohms). Then, 9×ZS (450 ohms) is transformed into ZS×9/4 (112 ohms) in the eighth transmission line transformer **183***c*. The same things hold true for the sixth input terminal **184***c*, the ninth transmission line transformer **185***c*, and the tenth transmission line transformer **186***c*. Then, at the combination terminal **188***c*, ZS×9/4 (112 ohms) is reduced by half. Hence, an impedance at the output terminal **189***c* is ZS×9/8 (about 50 ohms). Incidentally, in the eighth transmission line transformer **183***c* and the tenth transmission line transformer **186***c*, impedances may be finely adjusted.

Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

Fourth Form of Combiner **180***d*

A fourth form of combiner **180***d* will be described with reference to FIG. **7**. FIG. **7** illustrates an example of a configuration of the fourth form of combiner **180***d*.

As illustrated in FIG. **7**, the fourth form of combiner **180***d* includes, for example, a seventh input terminal **181***d*, an eleventh transmission line transformer **182***d*, a twelfth transmission line transformer **183***d*, an eighth input terminal **184***d*, a thirteenth transmission line transformer **185***d*, a fourteenth transmission line transformer **186***d*, a fourth isolation section **187***d*, a combination terminal **188***d*, and an output terminal **189***d*. The fourth form of combiner **180***d* is a combiner in which, in the third form of combiner **180***c*, the seventh and ninth transmission line transformers **182***c* and **185***c* with the transformation ratio of 1:9 are changed to the eleventh and thirteenth transmission line transformers **182***d* and **185***d* with a transformation ratio of 4:1 and the eighth and tenth transmission line transformers **183***c* and **186***c* with the transformation ratio of 4:1 are changed to the twelfth and fourteenth transmission line transformers **183***d* and **186***d* with a transformation ratio of 1:9. This enables low-loss and wideband power combination.

Fifth Form of Combiner **180***e*

A fifth form of combiner **180***e* will be described with reference to FIG. **8**. FIG. **8** illustrates an example of a configuration of the fifth form of combiner **180***e*.

As illustrated in FIG. **8**, the combiner **180***e* includes, for example, a ninth input terminal **181***e*, a tenth input terminal **182***e*, a fifteenth transmission line transformer **183***e*, a sixteenth transmission line transformer **184***e*, a fifth isolation section **185***e*, an output terminal **186***e*, and a combination terminal **187***e*.

The ninth input terminal **181***e* is a terminal to which an amplified signal RFamp**1** is inputted. The tenth input terminal **182***e* is a terminal to which an amplified signal RFamp**2** is inputted. The fifteenth transmission line transformer **183***e* is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the fifteenth transmission line transformer **183***e*, for example, the ninth input terminal **181***e* is connected to the second end portion **191***b* of the first transmission line **191** of the transmission line transformer **190** illustrated in FIG. **3**C, the tenth input terminal **182***e* is connected to the sixth end portion **193***b* (second end portion) of the third transmission line **193**, and the combination terminal **187***e* is connected to the first end portion **191***a* of the first transmission line **191**. Here, although FIG. **3**C illustrates the transmission line transformer **190** including the extended line **194**, the transmission line transformer **190** does not have to include the extended line **194**. In this case, the second end portion **191***b* of the first transmission line **191** is constructed so as to extend toward the outside of the first transmission line **191**, and the ninth input terminal **181***e* may be provided at the extended second end portion **191***b*. In the fifteenth transmission line transformer **183***e*, the other end is connected to one end of the sixteenth transmission line transformer **184***e* through the combination terminal **187***e*. The sixteenth transmission line transformer **184***e* is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:2.25. In the sixteenth transmission line transformer **184***e*, the one end is connected to the other end of the fifteenth transmission line transformer **183***e*, and the other end is connected to the output terminal **186***e*. The output terminal **186***e* is connected to the path-changing switch **170**.

Next, an overview of impedance transformation performed in the combiner **180***e* will be described. Each of impedances at the ninth and tenth input terminals **181***e* and **182***e* of the combiner **180***e* is ZS (50 ohms). Here, the fifteenth transmission line transformer **183***e* performs an operation of transforming the sum of impedances connected between the ninth input terminal **181***e* and the tenth input terminal **182***e* into a quarter. When powers in phase are inputted to the respective ninth and tenth input terminals **181***e* and **182***e* of the combiner **180***e*, an impedance between the ninth input terminal **181***e* and the tenth input terminal **182***e* can be regarded as a configuration in which two Zss are connected in series through a common terminal (ground) and is thus 100 ohms from 2×ZS. In the fifteenth transmission line transformer **183***e*, 2×ZS (100 ohms) is transformed into 2×ZS×¼ (25 ohms), and 25 ohms is reached at the combination terminal **187***e* (combination terminal). Then, 25 ohms is transformed into ZS/2×2.25 in the sixteenth transmission line transformer **184***e*, and an impedance at the output terminal **186***e* is ZS/2×2.25 (about 50 ohms). Incidentally, in the sixteenth transmission line transformer **184***e*, the impedance may be finely adjusted.

Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

Other Embodiments

A power amplifier module **200** according to another embodiment will be described with reference to FIGS. **9**A and **9**B. FIGS. **9**A and **9**B illustrate an overview of a configuration of the power amplifier module **200** according to the other embodiment.

The power amplifier module **200** according to the other embodiment includes a combiner **280** including the antenna switch **160** of the power amplifier module **100** illustrated in FIG. **1**. Other components are similar to those of the power amplifier module **100**, and thus a description thereof is omitted.

FIG. **9**A illustrates the power amplifier module **200** including a transmission line transformer between an antenna switch **282** and the antennas ANT**1** and ANT**2**. As illustrated in FIG. **9**A, the combiner **280** includes, for example, preceding transmission line transformers **281**, the antenna switch **282**, a subsequent transmission line transformer **283**, and a combination terminal **284**.

The preceding transmission line transformers **281** are connected to antennas ANT**1** and ANT**2** sides of duplexers

250. Each of the preceding transmission line transformers **281** performs impedance transformation at a predetermined transformation ratio. Specifically, the preceding transmission line transformers **281** correspond to, for example, in the power amplifier module **100**, the first and second transmission line transformers **182***a* and **184***a* in the combiner **180***a*, or the fourth and fifth transmission line transformers **182***b* and **184***b* in the combiner **180***b*.

Although the input terminals of the antenna switch **160** illustrated in FIG. **2** are connected to the duplexers **150**, input terminals of the antenna switch **282** are connected, for example, to the preceding transmission line transformers **281** in place of the duplexers **150**. The structure of the antenna switch **282** is similar to the structure of the antenna switch **160** illustrated in FIG. **2**, and thus a description thereof is omitted. In comparison with a combination terminal provided in the combiner **180** of the power amplifier module **100**, in the power amplifier module **200** according to the other embodiment, the combination terminal **284** is provided, for example, between the antenna switch **282** and the subsequent transmission line transformer **283**.

The subsequent transmission line transformer **283** is connected to an antennas ANT**1** and ANT**2** side of the antenna switch **282**. The subsequent transmission line transformer **283** performs impedance transformation at a predetermined transformation ratio. Specifically, the subsequent transmission line transformer **283** corresponds to, in the power amplifier module **100**, the third transmission line transformer **187***a* in the combiner **180***a*, or the sixth transmission line transformer **187***b* in the combiner **180***b*.

Thus, in comparison with the power amplifier module **1000** according to the comparative example, the power amplifier module **200** can be reduced in size.

FIG. **9**B illustrates the power amplifier module **200** including no transmission line transformer between the antenna switch **282** and the antennas ANT**1** and ANT**2**. As illustrated in FIG. **9**B, the combiner **280** includes, for example, the preceding transmission line transformers **281**, the antenna switch **282**, and the combination terminal **284**. The preceding transmission line transformers **281** are connected to the antennas ANT**1** and ANT**2** sides of the duplexers **250**. Each of the preceding transmission line transformers **281** performs impedance transformation at a predetermined transformation ratio. Specifically, the preceding transmission line transformers **281** correspond to, for example, in the power amplifier module **100**, the seventh and eighth transmission line transformers **182***c* and **183***c* and the ninth and tenth transmission line transformers **185***c* and **186***c* in the combiner **180***c*, or the eleventh and twelfth transmission line transformers **182***d* and **183***d* and the thirteenth and fourteenth transmission line transformers **185***d* and **186***d* in the combiner **180***d*. The antenna switch **282** is described above, and thus a description thereof is omitted. The combination terminal **284** is connected, for example, to a path-changing switch **270**. The path-changing switch **270** is similar to the path-changing switch **170**, and thus a description thereof is omitted. Thus, in comparison with the power amplifier module **1000** according to the comparative example, the power amplifier module **200** can be reduced in size.

Summary

The power amplifier module **100** according to the present embodiment includes the first amplifier **110***a* that amplifies a power level of an input signal RFin**1** (first input signal) in a predetermined frequency band and outputs an amplified signal RFamp**1** (first signal) of PC3 (first power level); the first impedance transformer **120***a* connected to the first amplifier **110***a* and including a transmission line transformer; the second amplifier **110***b* that amplifies a power level of an input signal RFin**2** (second input signal) in the predetermined frequency band and outputs an amplified signal RFamp**2** (second signal) of PC3 (first power level); the second impedance transformer **120***b* connected to the second amplifier **110***b* and including a transmission line transformer; and the combiner **180** that combines the amplified signal RFamp**1** (first signal) inputted through the first impedance transformer **120***a* and the amplified signal RFamp**2** (second signal) inputted through the second impedance transformer **120***b* into an output signal of PC2 (second power level) larger than PC3 (first power level) and includes a transmission line transformer. This enables an increase in transmission efficiency while achieving a reduction in module size.

Furthermore, the power amplifier module **100** according to the present embodiment further includes the antenna switch **160**. The antenna switch **160** includes one terminal (first terminal) of input terminals **161** to which the amplified signal RFamp**1** (first signal) is inputted through the first impedance transformer **120***a*, one terminal (second terminal) of input terminals **161** to which the amplified signal RFamp**2** (second signal) is inputted through the second impedance transformer **120***b*, the first output terminal **162***a* that outputs the amplified signal RFamp**1** (first signal) to the antenna ANT**1** (first antenna), the second output terminal **162***b* that outputs the amplified signal RFamp**2** (second signal) to the antenna ANT**2** (second antenna) different from the antenna ANT**1** (first antenna), and the third output terminals **162***c* that output the amplified signal RFamp**1** (first signal) and the amplified signal RFAmp**2** (second signal) to the combiner **180**. This enables an increase in transmission efficiency while achieving a reduction in module size. Furthermore, the combiner **180** is disposed on the antennas ANT**1** and ANT**2** side of the antenna switch **160**, and thus the impedances at the input and output terminals of the antenna switch **160** are about 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch **160**.

Furthermore, the first form of combiner **180***a* (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **186***a* at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined, the first transmission line transformer **182***a* that is connected between the combination terminal **186***a* and the first input terminal **181***a* to which the amplified signal RFamp**1** (first signal) is inputted and that performs impedance transformation at a transformation ratio of 1:9 (first transformation ratio) so that an impedance increases, the second transmission line transformer **184***a* that is connected between the combination terminal **186***a* and the second input terminal **183***a* to which the amplified signal RFamp**2** (second signal) is inputted and that performs impedance transformation at the transformation ratio of 1:9 (first transformation ratio) so that an impedance increases, the third transmission line transformer **187***a* that is connected between the combination terminal **186***a* and the output terminal **188***a* and that performs impedance transformation at a transformation ratio of 4:1 (second transformation ratio) so that an impedance decreases, and the first isolation section **185***a* connected between the first input terminal **181***a* and the second input terminal **183***a*. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

Furthermore, the second form of combiner **180***b* (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **186***b* at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined, the fourth transmission line transformer **182***b* that is connected between the combination terminal **186***b* and the third input terminal **181***b* to which the amplified signal RFamp**1** (first signal) is inputted and that performs impedance transformation at a transformation ratio of 4:1 (third transformation ratio) so that an impedance decreases, the fifth transmission line transformer **184***b* that is connected between the combination terminal **186***b* and the fourth input terminal **183***b* to which the amplified signal RFamp**2** (second signal) is inputted and that performs impedance transformation at the transformation ratio of 4:1 (third transformation ratio) so that an impedance decreases, the sixth transmission line transformer **187***b* that is connected between the combination terminal **186***b* and the output terminal **188***b* and that performs impedance transformation at a transformation ratio of 1:9 (fourth transformation ratio) so that an impedance increases, and the second isolation section **185***b* connected between the third input terminal **181***b* and the fourth input terminal **183***b*. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

Furthermore, the third form of combiner **180***c* (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **188***c* at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined, the seventh transmission line transformer **182***c* that performs impedance transformation at a transformation ratio of 1:9 (fifth transformation ratio) so that an impedance increases, the eighth transmission line transformer **183***c* that is connected in series with the seventh transmission line transformer **182***c* through the first connection terminal **1881***c* and that performs impedance transformation at a transformation ratio of 4:1 (sixth transformation ratio) so that the impedance decreases, the ninth transmission line transformer **185***c* that performs impedance transformation at the transformation ratio of 1:9 (fifth transformation ratio) so that an impedance increases, the tenth transmission line transformer **186***c* that is connected in series with the ninth transmission line transformer **185***c* through the second connection terminal **1882***c* and that performs impedance transformation at the transformation ratio of 4:1 (sixth transformation ratio) so that the impedance decreases, and the third isolation section **187***c*. The seventh transmission line transformer **182***c* is connected between the first connection terminal **1881***c* and the fifth input terminal **181***c* to which the amplified signal RFamp**1** (first signal) is inputted. The eighth transmission line transformer **183***c* is connected between the first connection terminal **1881***c* and the combination terminal **188***c*. The ninth transmission line transformer **185***c* is connected between the second connection terminal **1882***c* and the sixth input terminal **184***c* to which the amplified signal RFamp**2** (second signal) is inputted. The tenth transmission line transformer **186***c* is connected between the second connection terminal **1882***c* and the combination terminal **188***c*. The third isolation section **187***c* is connected between the fifth input terminal **181***c* and the sixth input terminal **184***c*. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

Furthermore, the fourth form of combiner **180***d* (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **188***d* at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined, the eleventh transmission line transformer **182***d* that performs impedance transformation at a transformation ratio of 4:1 (seventh transformation ratio) so that an impedance decreases, the twelfth transmission line transformer **183***d* that is connected in series with the eleventh transmission line transformer **182***d* through a third connection terminal **1881***d* and that performs impedance transformation at a transformation ratio of 1:9 (eighth transformation ratio) so that the impedance increases, the thirteenth transmission line transformer **185***d* that performs impedance transformation at the transformation ratio of 4:1 (seventh transformation ratio) so that an impedance decreases, the fourteenth transmission line transformer **186***d* that is connected in series with the thirteenth transmission line transformer **185***d* through a fourth connection terminal **1882***d* and that performs impedance transformation at the transformation ratio of 1:9 (eighth transformation ratio) so that the impedance increases, and the fourth isolation section **187***d*. The eleventh transmission line transformer **182***d* is connected between the third connection terminal **1881***d* and the seventh input terminal **181***d* to which the amplified signal RFamp**1** (first signal) is inputted. The twelfth transmission line transformer **183***d* is connected between the third connection terminal **1881***d* and the combination terminal **188***d*. The thirteenth transmission line transformer **185***d* is connected between the fourth connection terminal **1882***d* and the eighth input terminal **184***d* to which the amplified signal RFamp**2** (second signal) is inputted. The fourteenth transmission line transformer **186***d* is connected between the fourth connection terminal **1882***d* and the combination terminal **188***d*. The fourth isolation section **187***d* is connected between the seventh input terminal **181***d* and the eighth input terminal **184***d*. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

Furthermore, the fifth form of combiner **180***e* (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **187***e* at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined, the fifteenth transmission line transformer **183***e* that includes the second end portion **191***b* (first end portion) that is connected to the ninth input terminal **181***e* and to which the amplified signal RFamp**1** (first signal) is inputted, the sixth end portion **193***b* (second end portion) that is connected to the tenth input terminal **182***e* and to which the amplified signal RFamp**2** (second signal) is inputted, and the first end portion **191***a* (third end portion) connected to the combination terminal **187***e*, and that performs impedance transformation at a transformation ratio of 4:1 (ninth transformation ratio) so that an impedance decreases, the sixteenth transmission line transformer **184***e* that is connected in series with the fifteenth transmission line transformer **183***e* through the combination terminal **187***e* and connected between the combination terminal **187***e* and the output terminal **186***e*, and that performs impedance transformation at a transformation ratio of 1:2.25 (tenth transformation ratio) so that the impedance increases, and the fifth isolation section **185***e* connected between the ninth input terminal **181***e* and the tenth input terminal **182***e*. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

Furthermore, the combiner **280** of the power amplifier module **200** according to the present embodiment includes a preceding transmission line transformer **281** (seventeenth transmission line transformer) that is connected to a duplexer **150** (first duplexer) separating a reception signal and the amplified signal RFamp**1** (first signal) and that performs impedance transformation at a predetermined transformation ratio (eleventh transformation ratio), a preceding transmission line transformer **281** (eighteenth transmission line transformer) that is connected to a duplexer **150** (second duplexer) separating a reception signal and the amplified signal RFamp**2** (second signal) and that performs impedance transformation at the predetermined transformation ratio (eleventh transformation ratio), and the antenna switch **282** including an input terminal (eleventh input terminal) to which the amplified signal RFamp**1** (first signal) is inputted through the preceding transmission line transformer **281** (seventeenth transmission line transformer), an input terminal (twelfth input terminal) to which the amplified signal RFamp**2** (second signal) is inputted through the preceding transmission line transformer **281** (eighteenth transmission line transformer), a fourth output terminal connected to the antenna ANT**1** (first antenna), an output terminal (fifth output terminal) connected to the antenna ANT**2** (second antenna) different from the antenna ANT**1** (first antenna), and output terminals (sixth output terminals) connected to the combination terminal **284** at which the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) are combined. Thus, the power amplifier module **200** is reduced in size.

Furthermore, the combiner **280** of the power amplifier module **200** according to the present embodiment includes the subsequent transmission line transformer **283** (nineteenth transmission line transformer) that is connected between the combination terminal **284** and an output terminal (for example, a terminal connected to the path-changing switch **270**) and that performs impedance transformation at a predetermined transformation ratio (twelfth transformation ratio). Thus, the power amplifier module **200** can be reduced in size.

Furthermore, the combiner **180** or **280** in the power amplifier module **100** or **200** according to the present embodiment receives input of the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) through the first and second band selection switches **140***a* and **140***b* that distribute the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) in accordance with the predetermined frequency band. Thus, a module used in a wide band can be implemented.

Furthermore, the combiner **180** in the power amplifier module **100** according to the present embodiment receives input of the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal) through duplexers **150** that distribute the amplified signal RFamp**1** (first signal) and the amplified signal RFamp**2** (second signal), and a reception signal received. This enables transmission and reception using the same antenna, thus achieving a reduction in module size.

Furthermore, the predetermined frequency band of each input signal RFin in the power amplifier module **100** or **200** according to the present embodiment is a frequency band from 1710 MHz to 2690 MHz. When the first amplifier **110***a* and the second amplifier **110***b* that support a wide band are used, the number of components can be reduced, thus achieving a reduction in module size.

The above-described embodiments are intended to facilitate understanding of the present disclosure but are not intended for a limited interpretation of the present disclosure. The present disclosure can be changed or improved without departing from the gist thereof and also encompasses equivalents thereof. In other words, appropriate design changes made to the embodiments by those skilled in the art are also encompassed in the scope of the present disclosure as long as the changes have features of the present disclosure. The elements included in the embodiments, and the arrangement and so forth of the elements are not necessarily limited to those exemplified herein and can be appropriately changed. While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer;

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer; and

an antenna switch,

wherein the antenna switch includes:

a first terminal to which the first signal is inputted through the first impedance transformer,

a second terminal to which the second signal is inputted through the second impedance transformer,

a first output terminal configured to output the first signal to a first antenna,

a second output terminal configured to output the second signal to a second antenna different from the first antenna, and

third output terminals configured to output the first signal and the second signal to the combiner.

2. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a first transmission line transformer connected between the combination terminal and a first input terminal to which the first signal is inputted and configured to perform impedance transformation at a first transformation ratio so that an impedance increases, a second transmission line transformer connected between the combination terminal and a second input terminal to which the second signal is inputted and configured to perform impedance transformation at the first transformation ratio so that an impedance increases,

a third transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a second transformation ratio so that an impedance decreases, and

a first isolation section connected between the first input terminal and the second input terminal.

3. The power amplifier module according to claim **1**, wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a first transmission line transformer connected between the combination terminal and a first input terminal to which the first signal is inputted and configured to perform impedance transformation at a first transformation ratio so that an impedance increases,

a second transmission line transformer connected between the combination terminal and a second input terminal to which the second signal is inputted and configured to perform impedance transformation at the first transformation ratio so that an impedance increases,

a third transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a second transformation ratio so that an impedance decreases, and

a first isolation section connected between the first input terminal and the second input terminal.

4. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a fourth transmission line transformer connected between the combination terminal and a third input terminal to which the first signal is inputted and configured to perform impedance transformation at a third transformation ratio so that an impedance decreases,

a fifth transmission line transformer connected between the combination terminal and a fourth input terminal to which the second signal is inputted and configured to perform impedance transformation at the third transformation ratio so that an impedance decreases,

a sixth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a fourth transformation ratio so that an impedance increases, and

a second isolation section connected between the third input terminal and the fourth input terminal.

5. The power amplifier module according to claim **1**, wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a fourth transmission line transformer connected between the combination terminal and a third input terminal to which the first signal is inputted and configured to perform impedance transformation at a third transformation ratio so that an impedance decreases,

a fifth transmission line transformer connected between the combination terminal and a fourth input terminal to which the second signal is inputted and configured to perform impedance transformation at the third transformation ratio so that an impedance decreases,

a sixth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a fourth transformation ratio so that an impedance increases, and

a second isolation section connected between the third input terminal and the fourth input terminal.

6. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes;

a combination terminal at which the first signal and the second signal are combined,

a seventh transmission line transformer configured to perform impedance transformation at a fifth transformation ratio so that an impedance increases,

an eighth transmission line transformer connected in series with the seventh transmission line transformer through a first connection terminal and configured to perform impedance transformation at a sixth transformation ratio so that the impedance decreases,

a ninth transmission line transformer configured to perform impedance transformation at the fifth transformation ratio so that an impedance increases,

a tenth transmission line transformer connected in series with the ninth transmission line transformer through a second connection terminal and configured to perform impedance transformation at the sixth transformation ratio so that the impedance decreases, and

a third isolation section,

wherein the seventh transmission line transformer is connected between the first connection terminal and a fifth input terminal to which the first signal is inputted,

wherein the eighth transmission line transformer is connected between the first connection terminal and the combination terminal,

wherein the ninth transmission line transformer is connected between the second connection terminal and a sixth input terminal to which the second signal is inputted,

wherein the tenth transmission line transformer is connected between the second connection terminal and the combination terminal, and

wherein the third isolation section is connected between the fifth input terminal and the sixth input terminal.

7. The power amplifier module according to claim **1**,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a seventh transmission line transformer configured to perform impedance transformation at a fifth transformation ratio so that an impedance increases,

an eighth transmission line transformer connected in series with the seventh transmission line transformer through a first connection terminal and configured to perform impedance transformation at a sixth transformation ratio so that the impedance decreases,

a ninth transmission line transformer configured to perform impedance transformation at the fifth transformation ratio so that an impedance increases,

a tenth transmission line transformer connected in series with the ninth transmission line transformer through a second connection terminal and configured to perform impedance transformation at the sixth transformation ratio so that the impedance decreases, and

a third isolation section,

wherein the seventh transmission line transformer is connected between the first connection terminal and a fifth input terminal to which the first signal is inputted,

wherein the eighth transmission line transformer is connected between the first connection terminal and the combination terminal,

wherein the ninth transmission line transformer is connected between the second connection terminal and a sixth input terminal to which the second signal is inputted,

wherein the tenth transmission line transformer is connected between the second connection terminal and the combination terminal, and

wherein the third isolation section is connected between the fifth input terminal and the sixth input terminal.

8. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

an eleventh transmission line transformer configured to perform impedance transformation at a seventh transformation ratio so that an impedance decreases,

a twelfth transmission line transformer connected in series with the eleventh transmission line transformer through a third connection terminal and configured to perform impedance transformation at an eighth transformation ratio so that the impedance increases,

a thirteenth transmission line transformer configured to perform impedance transformation at the seventh transformation ratio so that an impedance decreases,

a fourteenth transmission line transformer connected in series with the thirteenth transmission line transformer through a fourth connection terminal and configured to perform impedance transformation at the eighth transformation ratio so that the impedance increases, and

a fourth isolation section,

wherein the eleventh transmission line transformer is connected between the third connection terminal and a seventh input terminal to which the first signal is inputted,

wherein the twelfth transmission line transformer is connected between the third connection terminal and the combination terminal,

wherein the thirteenth transmission line transformer is connected between the fourth connection terminal and an eighth input terminal to which the second signal is inputted,

wherein the fourteenth transmission line transformer is connected between the fourth connection terminal and the combination terminal, and

wherein the fourth isolation section is connected between the seventh input terminal and the eighth input terminal.

9. The power amplifier module according to claim **1**,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

an eleventh transmission line transformer configured to perform impedance transformation at a seventh transformation ratio so that an impedance decreases,

a twelfth transmission line transformer connected in series with the eleventh transmission line transformer through a third connection terminal and configured to perform impedance transformation at an eighth transformation ratio so that the impedance increases,

a thirteenth transmission line transformer configured to perform impedance transformation at the seventh transformation ratio so that an impedance decreases,

a fourteenth transmission line transformer connected in series with the thirteenth transmission line transformer through a fourth connection terminal and configured to perform impedance transformation at the eighth transformation ratio so that the impedance increases, and

a fourth isolation section,

wherein the eleventh transmission line transformer is connected between the third connection terminal and a seventh input terminal to which the first signal is inputted,

wherein the twelfth transmission line transformer is connected between the third connection terminal and the combination terminal,

wherein the thirteenth transmission line transformer is connected between the fourth connection terminal and an eighth input terminal to which the second signal is inputted,

wherein the fourteenth transmission line transformer is connected between the fourth connection terminal and the combination terminal, and

wherein the fourth isolation section is connected between the seventh input terminal and the eighth input terminal.

10. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a fifteenth transmission line transformer including a first end portion that is connected to a ninth input terminal and to which the first signal is inputted, a second end portion that is connected to a tenth input terminal and to which the second signal is inputted, and a third end portion connected to the combination terminal, and configured to perform impedance transformation at a ninth transformation ratio so that an impedance decreases,

a sixteenth transmission line transformer connected in series with the fifteenth transmission line transformer through the combination terminal and connected between the combination terminal and an output terminal, and configured to perform impedance transformation at a tenth transformation ratio so that the impedance increases, and

a fifth isolation section connected between the ninth input terminal and the tenth input terminal.

11. The power amplifier module according to claim **1**, wherein the combiner includes:

a combination terminal at which the first signal and the second signal are combined,

a fifteenth transmission line transformer including a first end portion that is connected to a ninth input terminal and to which the first signal is inputted, a second end portion that is connected to a tenth input terminal and to which the second signal is inputted, and a third end portion connected to the combination terminal, and configured to perform impedance transformation at a ninth transformation ratio so that an impedance decreases,

a sixteenth transmission line transformer connected in series with the fifteenth transmission line transformer through the combination terminal and connected between the combination terminal and an output terminal, and configured to perform impedance transformation at a tenth transformation ratio so that the impedance increases, and

a fifth isolation section connected between the ninth input terminal and the tenth input terminal.

12. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner includes;

a seventeenth transmission line transformer connected to a first duplexer that separates a reception signal and the first signal and configured to perform impedance transformation at an eleventh transformation ratio,

an eighteenth transmission line transformer connected to a second duplexer that separates a reception signal and the second signal and configured to perform impedance transformation at the eleventh transformation ratio, and

an antenna switch including an eleventh input terminal to which the first signal is inputted through the seventeenth transmission line transformer, a twelfth input terminal to which the second signal is inputted through the eighteenth transmission line transformer, a fourth output terminal connected to a first antenna, a fifth output terminal connected to a second antenna different from the first antenna, and sixth output terminals connected to a combination terminal at which the first signal and the second signal are combined.

13. The power amplifier module according to claim **12**, wherein the combiner includes a nineteenth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a twelfth transformation ratio.

14. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner receives input of the first signal and the second signal through band selection switches configured to distribute the first signal and the second signal in accordance with the predetermined frequency band.

15. A power amplifier module comprising:

a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level;

a first impedance transformer connected to the first amplifier and including a transmission line transformer;

a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level;

a second impedance transformer connected to the second amplifier and including a transmission line transformer; and

a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer,

wherein the combiner receives input of the first signal and the second signal through duplexers configured to distribute the first signal and the second signal, and a received reception signal.

16. The power amplifier module according to claim **1**, wherein the combiner receives input of the first signal and the second signal through duplexers configured to distribute the first signal and the second signal, and a received reception signal.

17. The power amplifier module according to claim **1**, wherein the predetermined frequency band is a frequency band from 1710 MHz to 2690 MHz.

18. The power amplifier module according to claim **2**, wherein the predetermined frequency band is a frequency band from 1710 MHz to 2690 MHz.

19. The power amplifier module according to claim **1**, wherein the combiner is connected between the antenna switch and the first antenna and the second antenna.

* * * * *